Supplementary Information

Limited concordance between storm-tide magnitude and standardized connected-lowland screening across Northwest Europe

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This Supplementary Information is generated from the same frozen source tables as the manuscript. Legacy degree-window, 15 m denominator and category-based tables are intentionally excluded.

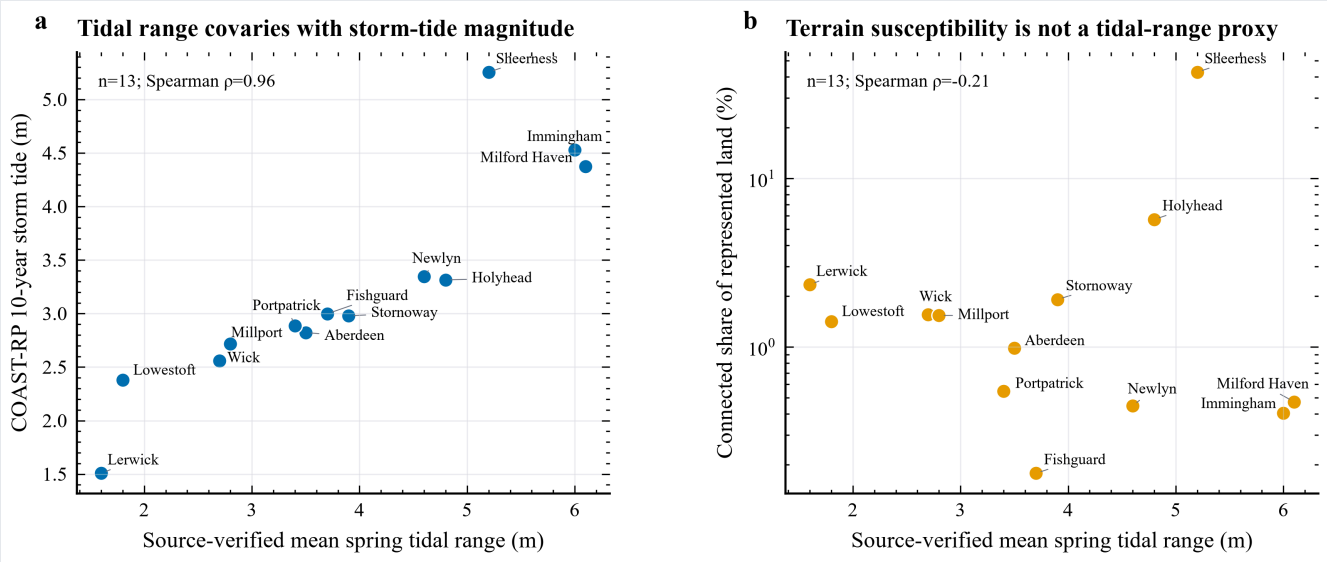
Supplementary Methods

The complete terrain grid contains 74 stations and 40 configurations per station (2,960 rows). The primary configuration is a 10 km window, standardized 2 m EGM2008 terrain threshold, eight-neighbour ocean-river seed, four-neighbour land graph, adaptive official-mask marine seed and COAST-RP match within 6 km. Only finite official-mask class-0 terrain enters the reference-land denominator. Internal outside-support value 254 and official classes 1, 2, 3 and 255 are excluded. Restricted permutation, block resampling and leave-one-block calculations use the versioned 100, 150 and 250 km identifiers in `config/coastal_spatial_blocks_v2.csv`.

The local terrain comparison retains each source product's native vertical datum. Continuous elevation was downscaled by the provider WCS; the service interpolation kernel was not controlled by the study code. Categorical masks and the connected mask used for intersection-over-union were reprojected by nearest neighbour to the local WCS grid.

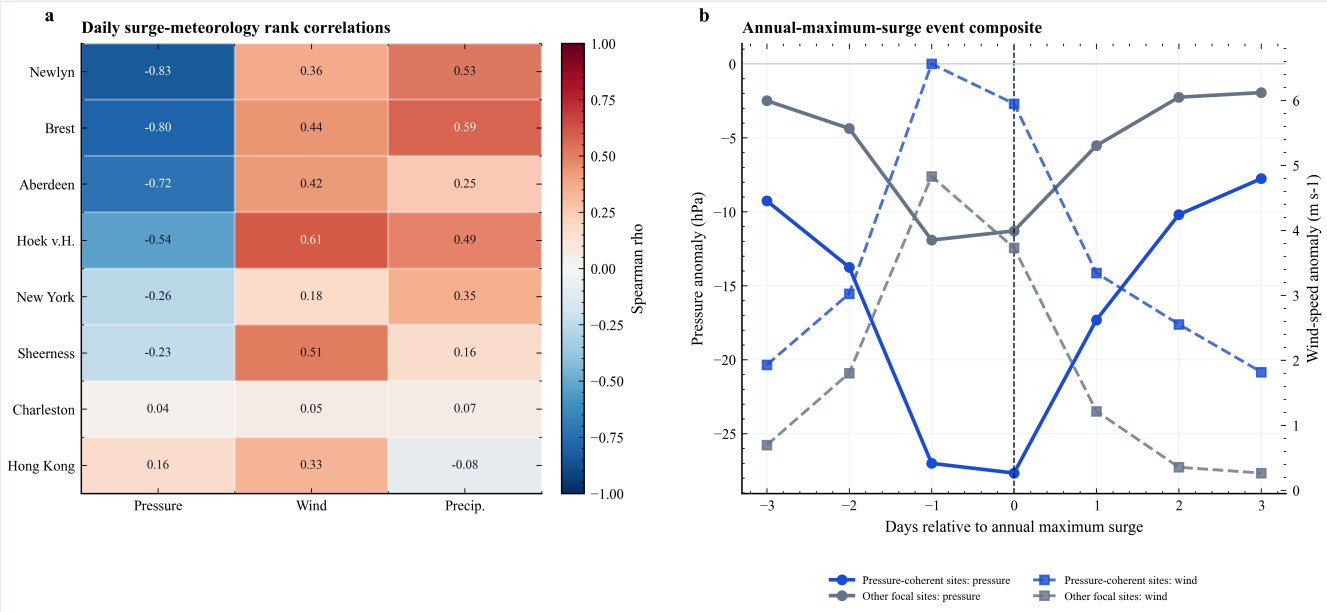
Supplementary Figures

Supplementary Figure S1. Source-verified tidal context



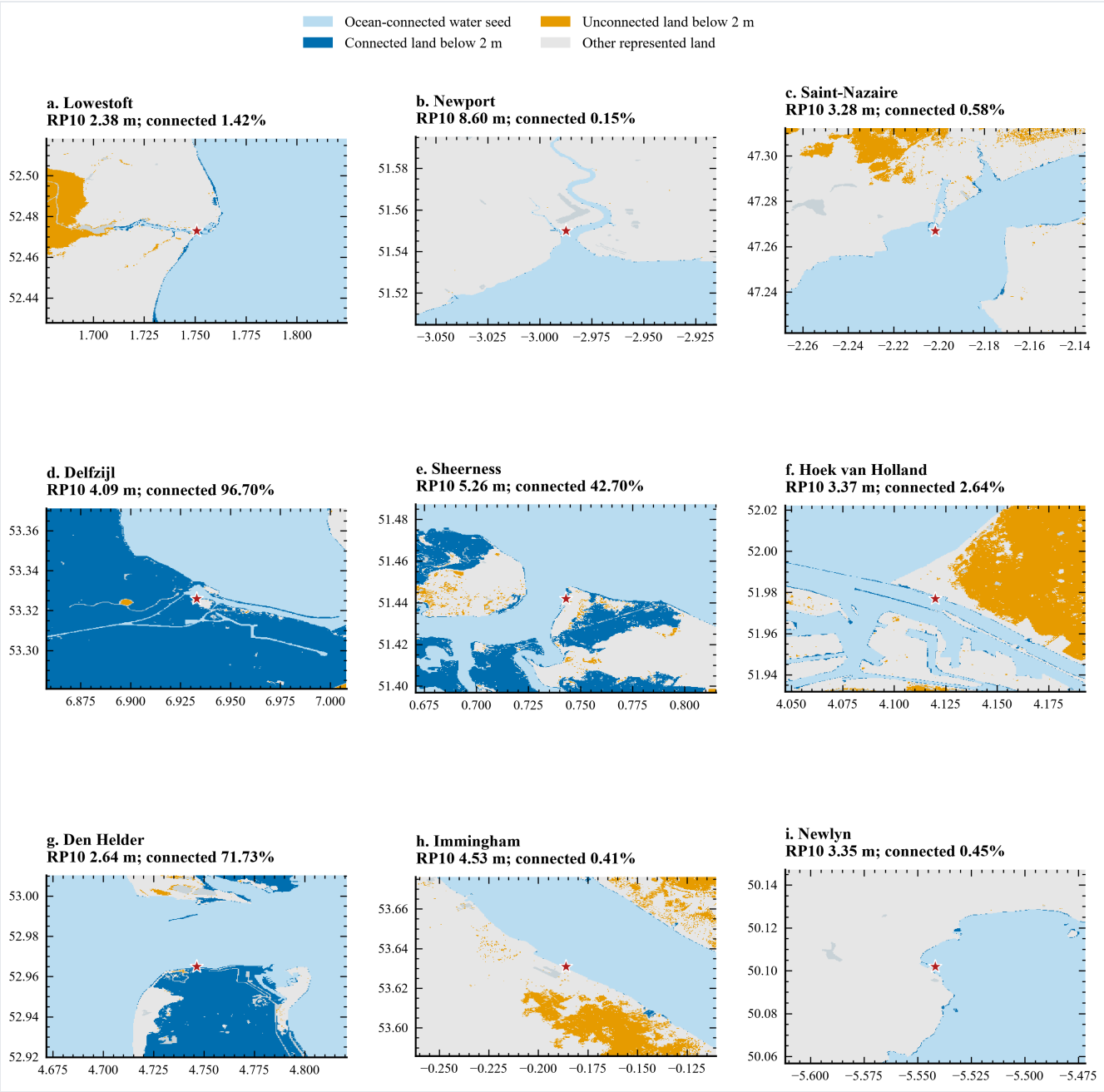
All 13 source-verified stations are plotted once. Tidal range covaries strongly with storm-tide magnitude but not with the standardized connected-terrain share in this small exploratory subset.

Supplementary Figure S2. Predictor-consistency diagnostic for focal GSSR series



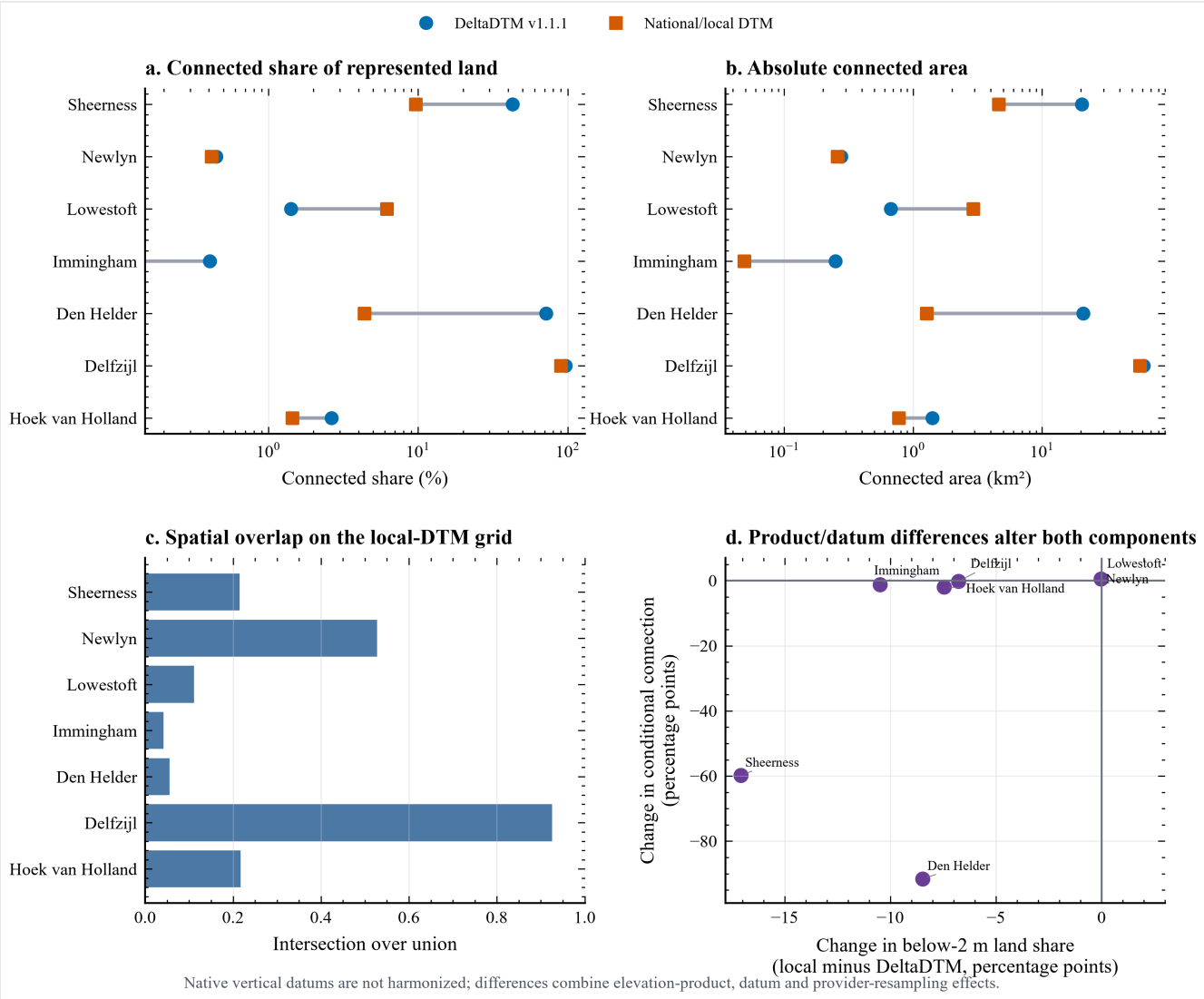
The atmospheric predictors overlap with the reconstruction framework. This panel is therefore not independent validation or extreme-event skill assessment and is not used in the terrain analysis.

Supplementary Figure S3. Standardized static terrain gallery



Every panel uses the same fixed 10 km window, official mask and 2 m EGM2008 threshold. Blue and orange cells retain their original raster footprint.

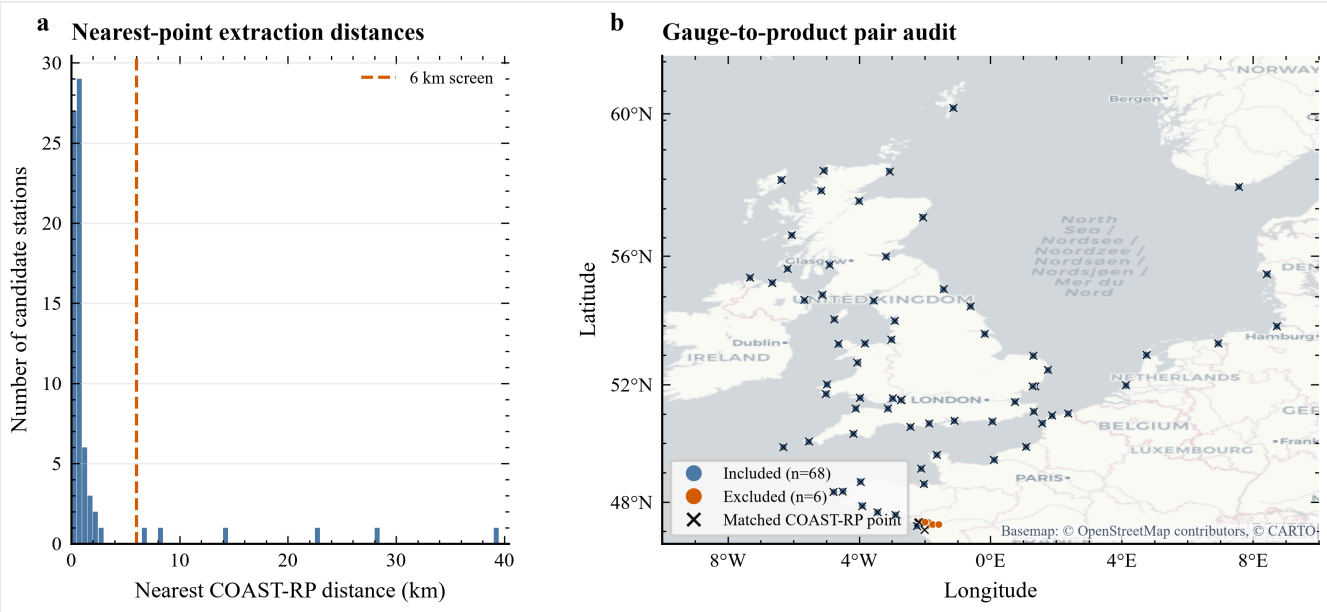
Supplementary Figure S4. Local-DTM product, datum and resampling sensitivity



Native vertical datums are not harmonized; differences combine elevation-product, datum and provider-resampling effects.

The three vertical references are not harmonized. The figure combines elevation-product, native-datum and provider-resampling effects and is not a validation result.

Supplementary Figure S5. COAST-RP nearest-point matching audit



All 74 candidate gauge-to-product pairs are retained in the source table. The six excluded Loire estuary stations lie 6.90-39.42 km from the nearest COAST-RP point.

Supplementary Tables

Supplementary Table S1. Frozen sample flow.

stage	n
GSSR metadata stations inside geographic bounds	74
DeltaDTM v1.1 elevation tiles available	74
Official marine seed resolved in 10 km window/context	74
COAST-RP nearest match within 6 km (primary)	68
GSSR record >=25 years and reconstruction correlation >=0.55	44

Supplementary Table S2. DeltaDTM v1.1.1 product-version and mask-fill audit.

product	release	tile tag	vertical reference	release cap m	paper v1 0 cap m	mask outside support fill	official 255 cells in primary analysis windows	outside support cells in primary analysis windows	note
DeltaDTM	v1.1.1	DeltaDTM v1.1.1 / DeltaDTM.jl v1.1.1	EGM2008	30.000	10.000	254	0	3320	The 2024 paper documents v1.0; the downloaded v1.1 README documents the 30 m EGM2008 cap. Outside-support fill is kept distinct from class 255 and excluded from all land denominators.

Supplementary Table S3. Primary association, spatial interval and top-set statistics.

Terrain metric	n	Block scale (km)	Number of blocks	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block empirical two-sided p	Top-set n	Observed overlap	Within-block null 2.5%	Within-block null 97.5%	Within-block lower-tail p	Block-resampling 2.5%	Block-resampling 97.5%
lowland_prevalence	68	100	42	-0.309	-0.188	-0.396	0.182	14	1	1.000	1.000	1.000	-0.549	0.033
lowland_prevalence	68	150	34	-0.309	-0.188	-0.477	0.022	14	1	1.000	2.000	0.333	-0.549	-0.007
lowland_prevalence	68	250	25	-0.309	-0.188	-0.417	0.020	14	1	1.000	2.000	0.332	-0.583	0.070
conditional_connectivity	68	100	42	0.105	0.069	0.231	0.490	14	5	4.000	5.000	1.000	-0.161	0.330
conditional_connectivity	68	150	34	0.105	0.069	0.139	0.585	14	5	3.000	6.000	0.917	-0.161	0.316
conditional_connectivity	68	250	25	0.105	0.069	0.147	0.464	14	5	2.000	6.000	0.971	-0.151	0.327

Terrain metric	n	Block scale (km)	Number of blocks	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block empirical two-sided p	Top-set n	Observed overlap	Within-block null 2.5%	Within-block null 97.5%	Within-block lower-tail p	Block-resampling 2.5%	Block-resampling 97.5%
connected_lowland_share	68	100	42	-0.333	-0.211	-0.350	0.135	14	1	1.000	1.000	1.000	-0.551	-0.027
connected_lowland_share	68	150	34	-0.333	-0.211	-0.455	0.025	14	1	1.000	2.000	0.332	-0.567	-0.027
connected_lowland_share	68	250	25	-0.333	-0.211	-0.412	0.012	14	1	1.000	4.000	0.099	-0.579	0.016
connected_area	68	100	42	-0.341	-0.208	-0.401	0.122	14	1	1.000	1.000	1.000	-0.566	-0.019
connected_area	68	150	34	-0.341	-0.208	-0.482	0.032	14	1	1.000	2.000	0.332	-0.559	-0.046
connected_area	68	250	25	-0.341	-0.208	-0.434	0.013	14	1	1.000	4.000	0.098	-0.577	0.016

Supplementary Table S4. COAST-RP return-period sensitivity with spatially constrained inference.

Return period (years)	Terrain metric	n	Block scale (km)	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block p	Top-20% overlap	Spatial-null lower-tail p
2	lowland_prevalence	68	100	-0.335	-0.205	-0.374	0.190	0	1.000
2	lowland_prevalence	68	150	-0.335	-0.205	-0.461	0.036	0	0.334
2	lowland_prevalence	68	250	-0.335	-0.205	-0.406	0.019	0	0.333
2	conditional_connectivity	68	100	0.104	0.066	0.219	0.499	5	1.000
2	conditional_connectivity	68	150	0.104	0.066	0.147	0.569	5	0.917
2	conditional_connectivity	68	250	0.104	0.066	0.163	0.416	5	0.932
2	connected_lowland_share	68	100	-0.359	-0.226	-0.326	0.156	0	0.333
2	connected_lowland_share	68	150	-0.359	-0.226	-0.435	0.028	0	0.113
2	connected_lowland_share	68	250	-0.359	-0.226	-0.394	0.018	0	0.046
2	connected_area	68	100	-0.369	-0.228	-0.376	0.149	0	0.335
2	connected_area	68	150	-0.369	-0.228	-0.464	0.040	0	0.112

Return period (years)	Terrain metric	n	Block scale (km)	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block p	Top-20% overlap	Spatial-null lower-tail p
2	connected_area	68	250	-0.369	-0.228	-0.418	0.019	0	0.047
2	connected_window_share	68	100	-0.369	-0.228	-0.376	0.156	0	0.337
2	connected_window_share	68	150	-0.369	-0.228	-0.464	0.041	0	0.111
2	connected_window_share	68	250	-0.369	-0.228	-0.418	0.017	0	0.048
5	lowland_prevalence	68	100	-0.318	-0.193	-0.384	0.196	1	1.000
5	lowland_prevalence	68	150	-0.318	-0.193	-0.469	0.034	1	0.338
5	lowland_prevalence	68	250	-0.318	-0.193	-0.409	0.022	1	0.336
5	conditional_connectivity	68	100	0.105	0.067	0.225	0.505	5	1.000
5	conditional_connectivity	68	150	0.105	0.067	0.139	0.581	5	0.919
5	conditional_connectivity	68	250	0.105	0.067	0.148	0.451	5	0.969
5	connected_lowland_share	68	100	-0.341	-0.216	-0.339	0.139	1	1.000
5	connected_lowland_share	68	150	-0.341	-0.216	-0.446	0.025	1	0.329
5	connected_lowland_share	68	250	-0.341	-0.216	-0.404	0.015	1	0.094
5	connected_area	68	100	-0.349	-0.215	-0.391	0.138	1	1.000
5	connected_area	68	150	-0.349	-0.215	-0.474	0.032	1	0.334
5	connected_area	68	250	-0.349	-0.215	-0.426	0.017	1	0.099
5	connected_window_share	68	100	-0.349	-0.215	-0.391	0.124	1	1.000
5	connected_window_share	68	150	-0.349	-0.215	-0.474	0.031	1	0.333
5	connected_window_share	68	250	-0.349	-0.215	-0.426	0.018	1	0.096
10	lowland_prevalence	68	100	-0.309	-0.188	-0.396	0.182	1	1.000
10	lowland_prevalence	68	150	-0.309	-0.188	-0.477	0.022	1	0.333
10	lowland_prevalence	68	250	-0.309	-0.188	-0.417	0.020	1	0.332
10	conditional_connectivity	68	100	0.105	0.069	0.231	0.490	5	1.000

Return period (years)	Terrain metric	n	Block scale (km)	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block p	Top-20% overlap	Spatial-null lower-tail p
10	conditional_connectivity	68	150	0.105	0.069	0.139	0.585	5	0.917
10	conditional_connectivity	68	250	0.105	0.069	0.147	0.464	5	0.971
10	connected_lowland_share	68	100	-0.333	-0.211	-0.350	0.135	1	1.000
10	connected_lowland_share	68	150	-0.333	-0.211	-0.455	0.025	1	0.332
10	connected_lowland_share	68	250	-0.333	-0.211	-0.412	0.012	1	0.099
10	connected_area	68	100	-0.341	-0.208	-0.401	0.122	1	1.000
10	connected_area	68	150	-0.341	-0.208	-0.482	0.032	1	0.332
10	connected_area	68	250	-0.341	-0.208	-0.434	0.013	1	0.098
10	connected_window_share	68	100	-0.341	-0.208	-0.401	0.114	1	1.000
10	connected_window_share	68	150	-0.341	-0.208	-0.482	0.037	1	0.334
10	connected_window_share	68	250	-0.341	-0.208	-0.434	0.016	1	0.092
25	lowland_prevalence	68	100	-0.304	-0.183	-0.396	0.176	1	1.000
25	lowland_prevalence	68	150	-0.304	-0.183	-0.472	0.030	1	0.338
25	lowland_prevalence	68	250	-0.304	-0.183	-0.418	0.013	1	0.331
25	conditional_connectivity	68	100	0.106	0.074	0.229	0.490	5	1.000
25	conditional_connectivity	68	150	0.106	0.074	0.125	0.607	5	0.914
25	conditional_connectivity	68	250	0.106	0.074	0.140	0.487	5	0.969
25	connected_lowland_share	68	100	-0.327	-0.206	-0.350	0.142	1	1.000
25	connected_lowland_share	68	150	-0.327	-0.206	-0.451	0.022	1	0.343
25	connected_lowland_share	68	250	-0.327	-0.206	-0.412	0.014	1	0.102
25	connected_area	68	100	-0.333	-0.204	-0.393	0.139	1	1.000
25	connected_area	68	150	-0.333	-0.204	-0.472	0.033	1	0.345
25	connected_area	68	250	-0.333	-0.204	-0.430	0.014	1	0.095

Return period (years)	Terrain metric	n	Block scale (km)	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block p	Top-20% overlap	Spatial-null lower-tail p
25	connected_window_share	68	100	-0.333	-0.204	-0.393	0.135	1	1.000
25	connected_window_share	68	150	-0.333	-0.204	-0.472	0.034	1	0.330
25	connected_window_share	68	250	-0.333	-0.204	-0.430	0.017	1	0.096
50	lowland_prevalence	68	100	-0.299	-0.180	-0.401	0.184	1	1.000
50	lowland_prevalence	68	150	-0.299	-0.180	-0.488	0.020	1	0.334
50	lowland_prevalence	68	250	-0.299	-0.180	-0.424	0.013	1	0.339
50	conditional_connectivity	68	100	0.100	0.068	0.223	0.507	5	1.000
50	conditional_connectivity	68	150	0.100	0.068	0.125	0.608	5	0.913
50	conditional_connectivity	68	250	0.100	0.068	0.130	0.532	5	0.968
50	connected_lowland_share	68	100	-0.323	-0.203	-0.355	0.137	1	1.000
50	connected_lowland_share	68	150	-0.323	-0.203	-0.467	0.017	1	0.334
50	connected_lowland_share	68	250	-0.323	-0.203	-0.420	0.011	1	0.093
50	connected_area	68	100	-0.328	-0.200	-0.393	0.145	1	1.000
50	connected_area	68	150	-0.328	-0.200	-0.481	0.026	1	0.326
50	connected_area	68	250	-0.328	-0.200	-0.434	0.012	1	0.094
50	connected_window_share	68	100	-0.328	-0.200	-0.393	0.125	1	1.000
50	connected_window_share	68	150	-0.328	-0.200	-0.481	0.030	1	0.332
50	connected_window_share	68	250	-0.328	-0.200	-0.434	0.013	1	0.095
100	lowland_prevalence	68	100	-0.290	-0.177	-0.426	0.153	1	1.000
100	lowland_prevalence	68	150	-0.290	-0.177	-0.498	0.019	1	0.341
100	lowland_prevalence	68	250	-0.290	-0.177	-0.428	0.013	1	0.342
100	conditional_connectivity	68	100	0.095	0.066	0.207	0.541	5	1.000
100	conditional_connectivity	68	150	0.095	0.066	0.107	0.689	5	0.921

Return period (years)	Terrain metric	n	Block scale (km)	Spearman rho	Kendall tau	Block-adjusted rank association	Within-block p	Top-20% overlap	Spatial-null lower-tail p
100	conditional_connectivity	68	250	0.095	0.066	0.119	0.568	5	0.969
100	connected_lowland_share	68	100	-0.317	-0.198	-0.389	0.093	1	1.000
100	connected_lowland_share	68	150	-0.317	-0.198	-0.487	0.013	1	0.336
100	connected_lowland_share	68	250	-0.317	-0.198	-0.430	0.007	1	0.094
100	connected_area	68	100	-0.322	-0.196	-0.417	0.117	1	1.000
100	connected_area	68	150	-0.322	-0.196	-0.494	0.023	1	0.342
100	connected_area	68	250	-0.322	-0.196	-0.439	0.013	1	0.089
100	connected_window_share	68	100	-0.322	-0.196	-0.417	0.109	1	1.000
100	connected_window_share	68	150	-0.322	-0.196	-0.494	0.023	1	0.333
100	connected_window_share	68	250	-0.322	-0.196	-0.439	0.014	1	0.095

Supplementary Table S5. Full 68-station primary source table.

Station ID	Station	Coastal basin	150 km spatial block	Latitude	Longitude	COAST-RP RP10 (m)	Match distance (km)	Below-2 m land share (%)	Conditional connection (%)	Connected land share (%)	Connected area (km2)
avonmouth-p060-uk	Avonmouth P060 Uk	Bristol-Channel	B150_UKBC02	51.511	-2.715	8.787	0.138	0.455	97.123	0.442	0.240
portbury-p341-uk	Portbury P341 Uk	Bristol-Channel	B150_UKBC02	51.500	-2.728	8.738	0.412	0.383	97.611	0.374	0.218
newport-p057-uk	Newport P057 Uk	Bristol-Channel	B150_UKBC02	51.550	-2.987	8.602	0.711	0.164	90.528	0.148	0.102
hinkleypoint-p003-uk	Hinkleypoint P003 Uk	Bristol-Channel	B150_UKBC02	51.215	-3.134	7.734	0.608	0.109	100.000	0.109	0.043
saint-malo	Saint Malo France	French-English-Channel	B150_FRC01	48.641	-2.028	7.428	0.129	0.487	68.459	0.333	0.183
heysham-p050-uk	Heysham P050 Uk	Irish-Sea	B150_UKI03	54.032	-2.920	6.624	0.153	1.124	99.103	1.114	0.371
sthelier-p074-uk	Sthelier P074 Uk	French-English-Channel	B150_FRC02	49.183	-2.117	6.501	0.113	0.228	100.000	0.228	0.125
dieppe	Dieppe France	French-English-Channel	B150_FRC03	49.929	1.085	6.143	0.537	0.288	85.823	0.248	0.134
mumbles-p932-uk	Mumbles P932 Uk	Bristol-Channel	B150_UKBC03	51.570	-3.975	5.825	0.111	0.207	93.830	0.195	0.045

Station ID	Station	Coastal basin	150 km spatial block	Latitude	Longitude	COAST-RP RP10 (m)	Match distance (km)	Below-2 m land share (%)	Conditional connection (%)	Connected land share (%)	Connected area (km2)
ilfracombe-p061-uk	Ilfracombe P061 Uk	Bristol-Channel	B150_UKBC01	51.211	-4.111	5.765	0.789	0.171	100.000	0.171	0.082
boulogne-sur-mer	Boulogne Sur Mer	French-English-Channel	B150_FRC04	50.727	1.578	5.713	0.801	0.944	99.795	0.942	0.441
gladstone-p234-uk	Gladstone P234 Uk	Irish-Sea	B150_UKI03	53.450	-3.018	5.577	0.724	0.309	66.396	0.205	0.139
workington-p217-uk	Workington P217 Uk	Irish-Sea	B150_UKI03	54.651	-3.568	5.426	1.113	0.819	100.000	0.819	0.362
sheerness-p015-uk	Sheerness	Thames-Estuary	B150_UKSE01	51.442	0.743	5.256	0.493	45.712	93.412	42.700	20.364
llandudno-p007-uk	Llandudno P007 Uk	Irish-Sea	B150_UKI02	53.332	-3.825	5.041	0.588	0.826	42.320	0.349	0.106
roscoff	Roscoff France	French-English-Channel	B150_FRC01	48.718	-3.966	4.857	0.619	0.911	95.320	0.869	0.244
calais	Calais France	French-English-Channel	B150_FRC04	50.969	1.868	4.700	1.668	37.398	76.721	28.692	14.955
cuxhaven-germany	Cuxhaven Germany	German-Bight	B150_DE01	53.867	8.717	4.696	0.531	61.332	97.032	59.512	28.550
le-havre	Le Havre France	French-English-Channel	B150_FRC03	49.482	0.106	4.643	1.113	3.597	92.096	3.312	1.401
dover-p012-uk	Dover P012 Uk	English-Channel-UK	B150_UKC03	51.117	1.318	4.618	0.508	0.285	93.878	0.268	0.124
dunkerque	Dunkerque France	French-English-Channel	B150_FRC04	51.048	2.367	4.548	1.002	34.580	88.407	30.571	16.083
immingham-p026-uk	Immingham P026 Uk	Southern-North-Sea-UK	B150_UKNE02	53.631	-0.186	4.528	0.487	20.142	2.014	0.406	0.249
milfordhaven-p056-uk	Milfordhaven P056 Uk	Celtic-Irish-Sea	B150_UKI01	51.702	-5.014	4.376	0.186	0.573	82.337	0.472	0.367
brest	Brest	Brittany-Atlantic	B150_FRB02	48.383	-4.500	4.313	0.833	0.459	61.245	0.281	0.176
newhaven-p011-uk	Newhaven P011 Uk	English-Channel-UK	B150_UKC03	50.782	0.057	4.261	0.540	5.849	64.710	3.785	2.031
delfzijl-del-nl	Delfzijl Del Nl	Dutch-German-Bight	B150_DE01	53.326	6.933	4.094	0.294	97.003	99.692	96.704	61.303
le-conquet	Le Conquet France	Brittany-Atlantic	B150_FRB02	48.359	-4.781	4.027	0.374	0.338	90.053	0.304	0.109

Station ID	Station	Coastal basin	150 km spatial block	Latitude	Longitude	COAST-RP RP10 (m)	Match distance (km)	Below-2 m land share (%)	Conditional connection (%)	Connected land share (%)	Connected area (km2)
harwich-p022-uk	Harwich P022 Uk	Southern-North-Sea-UK	B150_UKSE02	51.947	1.284	3.861	0.477	24.490	90.941	22.272	12.966
felixstowe-p204-uk	Felixstowe P204 Uk	Southern-North-Sea-UK	B150_UKSE02	51.957	1.348	3.750	0.806	27.837	96.247	26.793	10.628
cromer-p183-uk	Cromer P183 Uk	Southern-North-Sea-UK	B150_UKNE01	52.934	1.302	3.740	0.436	0.315	100.000	0.315	0.142
cherbourg	Cherbourg France	French-English-Channel	B150_FRC02	49.651	-1.635	3.674	0.180	1.470	91.840	1.351	0.779
porterin-p919-uk	Porterin P919 Uk	Irish-Sea	B150_UKI03	54.085	-4.768	3.640	0.445	2.170	100.000	2.170	0.723
leith-p034-uk	Leith P034 Uk	Northern-North-Sea	B150_UKNE04	55.990	-3.182	3.589	0.519	0.293	82.824	0.243	0.088
barmouth-p923-uk	Barmouth P923 Uk	Irish-Sea	B150_UKI02	52.719	-4.045	3.525	1.544	6.693	87.946	5.886	3.535
whitby-p174-uk	Whitby P174 Uk	Northern-North-Sea	B150_UKNE03	54.490	-0.614	3.431	0.684	0.449	61.040	0.274	0.135
esbjerg-130121-denmark	Esbjerg 130121 Denmark	Danish-Wadden-Coast	B150_DK01	55.467	8.433	3.399	1.096	9.225	88.797	8.192	4.926
hoekvanholla-hvh-nl	Hoek van Holland	Rhine-Meuse-Coast	B150_NL01	51.977	4.120	3.373	0.432	48.135	5.484	2.640	1.411
stmarys-p231-uk	Stmarys P231 Uk	Southwest-Atlantic	B150_UKSW01	49.918	-6.316	3.371	0.469	18.971	96.030	18.218	2.911
newlyn-p001-uk	Newlyn	Southwest-Atlantic	B150_UKSW01	50.102	-5.542	3.346	0.899	0.454	98.367	0.446	0.276
holyhead-p054-uk	Holyhead P054 Uk	Irish-Sea	B150_UKI02	53.308	-4.631	3.315	0.641	7.956	71.688	5.703	2.190
northshields-p032-uk	Northshields P032 Uk	Northern-North-Sea	B150_UKNE04	55.007	-1.440	3.302	0.936	0.199	76.421	0.152	0.103
saint-nazaire	Saint Nazaire France	Loire-Bay-of-Biscay	B150_FRL01	47.267	-2.202	3.275	0.513	10.684	5.465	0.584	0.331
ullapool-p043-uk	Ullapool P043 Uk	Scottish-Atlantic	B150_UKWA03	57.895	-5.158	3.256	0.491	0.279	98.970	0.276	0.219
saint-gildas	Saint Gildas France	Loire-Bay-of-Biscay	B150_FRL01	47.140	-2.246	3.243	0.696	4.497	99.308	4.466	0.652
kinlochbervi-p918-uk	Kinlochbervi P918 Uk	Scottish-Atlantic	B150_UKWA03	58.457	-5.050	3.151	2.314	0.730	94.512	0.690	0.490
le-crouesty	Le Crouesty France	Loire-Bay-of-Biscay	B150_FRB01	47.543	-2.895	3.131	0.199	4.551	35.605	1.620	0.439

Station ID	Station	Coastal basin	150 km spatial block	Latitude	Longitude	COAST-RP RP10 (m)	Match distance (km)	Below-2 m land share (%)	Conditional connection (%)	Connected land share (%)	Connected area (km2)
tobermory-p223-uk	Tobermory P223 Uk	Scottish-Atlantic	B150_UKWA02	56.623	-6.064	3.091	0.454	0.530	99.448	0.527	0.284
devonport-p002-uk	Devonport P002 Uk	Southwest-Atlantic	B150_UKSW01	50.368	-4.185	3.048	0.253	0.980	95.605	0.937	0.615
concarneau	Concarneau France	Brittany-Atlantic	B150_FRB01	47.874	-3.907	3.035	1.373	0.955	98.821	0.944	0.644
port-tudy	Port Tudy France	Brittany-Atlantic	B150_FRB01	47.644	-3.446	3.000	0.307	1.247	99.666	1.243	0.192
fishguard-p055-uk	Fishguard P055 Uk	Celtic-Irish-Sea	B150_UKI01	52.014	-4.983	2.998	0.642	0.227	78.340	0.178	0.072
morayfirth-p207-uk	Morayfirth P207 Uk	Northern-North-Sea	B150_UKNE05	57.599	-4.002	2.990	0.704	4.139	92.637	3.834	1.448
stornoway-p042-uk	Stornoway P042 Uk	Scottish-Atlantic	B150_UKWA03	58.208	-6.388	2.980	1.965	2.262	84.451	1.911	1.343
portpatrick-p063-uk	Portpatrick P063 Uk	North-Channel-Clyde	B150_UKI04	54.843	-5.120	2.887	0.779	0.548	99.669	0.546	0.248
aberdeen-p038-uk	Aberdeen	Northern-North-Sea	B150_UKNE05	57.143	-2.075	2.822	1.264	1.170	83.981	0.982	0.523
portsmouth-p008-uk	Portsmouth P008 Uk	English-Channel-UK	B150_UKC02	50.803	-1.112	2.767	0.689	16.896	88.826	15.008	9.510
millport-p049-uk	Millport P049 Uk	North-Channel-Clyde	B150_UKI04	55.750	-4.906	2.717	0.664	1.816	84.633	1.537	0.688
denhelder-hel-nl	Denhelder Hel Nl	Dutch-Wadden-Coast	B150_NL02	52.965	4.746	2.636	0.627	72.950	98.327	71.730	20.855
bangor-p662-uk	Bangor P662 Uk	North-Channel-Clyde	B150_UKI04	54.665	-5.670	2.615	2.811	1.592	100.000	1.592	0.913
wick-p035-uk	Wick P035 Uk	Northern-North-Sea	B150_UKNE05	58.441	-3.086	2.560	2.044	1.580	98.213	1.552	0.864
malin-head-834a-ireland	Malin Head 834a	Scottish-Atlantic	B150_UKWA01	55.367	-7.333	2.555	0.344	1.191	99.421	1.184	0.419
lowestoft-p024-uk	Lowestoft P024 Uk	Southern-North-Sea-UK	B150_UKNE01	52.473	1.751	2.379	0.559	12.097	11.707	1.416	0.670
portrush-p935-uk	Portrush P935 Uk	Scottish-Atlantic	B150_UKWA01	55.207	-6.657	2.005	0.451	1.115	82.627	0.922	0.389
weymouth-p991-uk	Weymouth P991 Uk	English-Channel-UK	B150_UKC01	50.608	-2.448	1.902	0.839	5.139	59.202	3.043	1.350
lerwick-p041-uk	Lerwick P041 Uk	Northern-North-Sea	B150_UKNE06	60.150	-1.133	1.508	0.094	2.493	93.779	2.338	1.331

Station ID	Station	Coastal basin	150 km spatial block	Latitude	Longitude	COAST-RP RP10 (m)	Match distance (km)	Below-2 m land share (%)	Conditional connection (%)	Connected land share (%)	Connected area (km2)
portellen-p202-uk	Portellen P202 Uk	Scottish-Atlantic	B150_UKWA01	55.627	-6.190	1.437	0.069	0.963	99.287	0.956	0.562
bournemouth-p988-uk	Bournemouth P988 Uk	English-Channel-UK	B150_UKC01	50.714	-1.875	1.395	0.825	0.883	81.183	0.717	0.375
tregde-006-norway	Tregde 006 Norway	Skagerrak	B150_NO01	58.000	7.567	0.999	0.035	6.605	86.397	5.707	2.743

Supplementary Table S6. Window support and alternative connected-window denominator.

Station	Nominal window area (km2)	Represented land area (km2)	Represented land (%)	Below-2 m area (km2)	Connected area (km2)	Connected window share (%)
Aberdeen	100.569	53.285	52.983	0.623	0.523	0.520
Avonmouth P060 Uk	100.461	54.457	54.207	0.248	0.240	0.239
Bangor P662 Uk	100.642	57.315	56.950	0.913	0.913	0.907
Barmouth P923 Uk	100.601	60.045	59.687	4.019	3.535	3.513
Boulogne Sur Mer	100.703	46.826	46.499	0.442	0.441	0.438
Bournemouth P988 Uk	100.436	52.338	52.111	0.462	0.375	0.374
Brest	100.595	62.636	62.266	0.288	0.176	0.175
Calais France	100.672	52.124	51.776	19.493	14.955	14.856
Cherbourg France	100.481	57.661	57.384	0.848	0.779	0.775
Concarneau France	100.554	68.207	67.831	0.652	0.644	0.640
Cromer P183 Uk	100.666	45.204	44.905	0.142	0.142	0.141
Cuxhaven Germany	100.683	47.974	47.648	29.423	28.550	28.356
Delfzijl Del Nl	100.495	63.392	63.080	61.492	61.303	61.001
Denhelder Hel Nl	100.594	29.074	28.902	21.209	20.855	20.731
Devonport P002 Uk	100.581	65.644	65.265	0.644	0.615	0.612
Dieppe France	100.507	54.076	53.803	0.156	0.134	0.133
Dover P012 Uk	100.448	46.321	46.114	0.132	0.124	0.123
Dunkerque France	100.599	52.607	52.294	18.192	16.083	15.987
Esbjerg 130121 Denmark	100.496	60.130	59.833	5.547	4.926	4.901
Felixstowe P204 Uk	100.624	39.669	39.423	11.043	10.628	10.563
Fishguard P055 Uk	100.496	40.629	40.428	0.092	0.072	0.072
Gladstone P234 Uk	100.573	67.842	67.455	0.210	0.139	0.138
Harwich P022 Uk	100.646	58.218	57.845	14.258	12.966	12.883
Heysham P050 Uk	100.652	33.354	33.138	0.375	0.371	0.369
Hinkleypoint P003 Uk	100.527	39.514	39.307	0.043	0.043	0.043
Hoek van Holland	100.579	53.453	53.145	25.730	1.411	1.403
Holyhead P054 Uk	100.631	38.395	38.155	3.055	2.190	2.176

Station	Nominal window area (km2)	Represented land area (km2)	Represented land (%)	Below-2 m area (km2)	Connected area (km2)	Connected window share (%)
Ilfracombe P061 Uk	100.536	48.265	48.007	0.082	0.082	0.082
Immingham P026 Uk	100.698	61.511	61.085	12.389	0.249	0.248
Kinlochbervi P918 Uk	100.552	71.053	70.664	0.519	0.490	0.488
Le Conquet France	100.643	35.849	35.620	0.121	0.109	0.108
Le Crouesty France	100.565	27.100	26.947	1.233	0.439	0.437
Le Havre France	100.627	42.298	42.034	1.521	1.401	1.392
Leith P034 Uk	100.551	36.078	35.880	0.106	0.088	0.087
Lerwick P041 Uk	100.621	56.926	56.574	1.419	1.331	1.323
Llandudno P007 Uk	100.574	30.347	30.174	0.251	0.106	0.105
Lowestoft P024 Uk	100.599	47.276	46.995	5.719	0.670	0.666
Malin Head 834a	100.486	35.379	35.208	0.421	0.419	0.417
Milfordhaven P056 Uk	100.617	77.703	77.226	0.445	0.367	0.364
Millport P049 Uk	100.561	44.739	44.490	0.813	0.688	0.684
Morayfirth P207 Uk	100.660	37.755	37.507	1.563	1.448	1.438
Mumbles P932 Uk	100.524	23.148	23.027	0.048	0.045	0.045
Newhaven P011 Uk	100.487	53.674	53.414	3.139	2.031	2.022
Newlyn	100.544	61.934	61.599	0.281	0.276	0.275
Newport P057 Uk	100.568	68.877	68.488	0.113	0.102	0.101
Northshields P032 Uk	100.596	67.751	67.350	0.135	0.103	0.102
Port Tudy France	100.581	15.418	15.329	0.192	0.192	0.190
Portbury P341 Uk	100.485	58.347	58.066	0.224	0.218	0.217
Portellen P202 Uk	100.615	58.849	58.490	0.566	0.562	0.559
Porterin P919 Uk	100.705	33.311	33.078	0.723	0.723	0.718
Portpatrick P063 Uk	100.469	45.421	45.209	0.249	0.248	0.247
Portrush P935 Uk	100.625	42.190	41.928	0.471	0.389	0.386
Portsmouth P008 Uk	100.540	63.365	63.024	10.706	9.510	9.459
Roscoff France	100.547	28.040	27.887	0.256	0.244	0.242
Saint Gildas France	100.489	14.602	14.531	0.657	0.652	0.649
Saint Malo France	100.496	55.045	54.774	0.268	0.183	0.182
Saint Nazaire France	100.671	56.653	56.275	6.053	0.331	0.329
Sheerness	100.613	47.691	47.401	21.801	20.364	20.240
Sthelier P074 Uk	100.631	55.017	54.672	0.125	0.125	0.125
Stmarys P231 Uk	100.530	15.979	15.895	3.031	2.911	2.896
Stornoway P042 Uk	100.525	70.287	69.920	1.590	1.343	1.336
Tobermory P223 Uk	100.610	53.852	53.526	0.285	0.284	0.282
Tregde 006 Norway	100.537	48.073	47.816	3.175	2.743	2.729
Ullapool P043 Uk	100.665	79.382	78.857	0.221	0.219	0.218

Station	Nominal window area (km2)	Represented land area (km2)	Represented land (%)	Below-2 m area (km2)	Connected area (km2)	Connected window share (%)
Weymouth P991 Uk	100.663	44.385	44.093	2.281	1.350	1.342
Whitby P174 Uk	100.533	49.430	49.168	0.222	0.135	0.135
Wick P035 Uk	100.597	55.683	55.353	0.880	0.864	0.859
Workington P217 Uk	100.677	44.181	43.884	0.362	0.362	0.359

Supplementary Table S7. Denominator-adjusted associations.

terrain metric	terrain metric label	n sites	raw spearman water vs metric	partial spearman controlling represented land fraction	spearman metric vs represented land fraction	spearman water vs represented land fraction
below_reference_land_pct	lowland_prevalence	68	-0.309	-0.310	-0.098	0.003
conditional_connectivity_pct	conditional_connectivity	68	0.105	0.106	-0.128	0.003
connected_reference_land_pct	connected_lowland_share	68	-0.333	-0.336	-0.137	0.003
connected_area_km2	connected_area	68	-0.341	-0.342	0.051	0.003
connected_window_pct	connected_window_share	68	-0.341	-0.342	0.051	0.003

Supplementary Table S8. One-at-a-time full-sample robustness grid.

Setting	Terrain metric	n	Spearman rho	Kendall tau	Two-sided p	Terrain-rank rho vs primary	Top-20% Jaccard vs primary	Top-20% membership changes
primary:10km,2m,4n,adaptive,match<=6km	connected_reference_land_pct	68	-0.333	-0.211	0.006	1.000	1.000	0
primary:10km,2m,4n,adaptive,match<=6km	below_reference_land_pct	68	-0.309	-0.188	0.009	1.000	1.000	0
primary:10km,2m,4n,adaptive,match<=6km	conditional_connectivity_pct	68	0.105	0.069	0.394	1.000	1.000	0
primary:10km,2m,4n,adaptive,match<=6km	connected_area_km2	68	-0.341	-0.208	0.005	1.000	1.000	0
window:5km	connected_reference_land_pct	68	-0.410	-0.273	6.999e-04	0.895	0.556	8
window:5km	below_reference_land_pct	68	-0.333	-0.217	0.006	0.932	0.647	6
window:5km	conditional_connectivity_pct	68	-0.003	-0.023	0.984	0.747	0.400	12
window:5km	connected_area_km2	68	-0.425	-0.279	2.000e-04	0.898	0.556	8

Setting	Terrain metric	n	Spearman rho	Kendall tau	Two-sided p	Terrain-rank rho vs primary	Top-20% Jaccard vs primary	Top-20% membership changes
window:20km	connected_reference_land_pct	68	-0.388	-0.267	0.001	0.910	0.750	4
window:20km	below_reference_land_pct	68	-0.317	-0.212	0.010	0.916	1.000	0
window:20km	conditional_connectivity_pct	68	0.032	0.019	0.799	0.665	0.400	12
window:20km	connected_area_km2	68	-0.407	-0.277	6.999e-04	0.890	0.647	6
threshold:1m	connected_reference_land_pct	68	-0.418	-0.271	3.000e-04	0.844	0.750	4
threshold:1m	below_reference_land_pct	68	-0.360	-0.239	0.002	0.950	0.867	2
threshold:1m	conditional_connectivity_pct	68	0.045	0.029	0.714	0.783	0.647	6
threshold:1m	connected_area_km2	68	-0.422	-0.273	4.000e-04	0.844	0.750	4
threshold:3m	connected_reference_land_pct	68	-0.295	-0.177	0.014	0.938	0.750	4
threshold:3m	below_reference_land_pct	68	-0.238	-0.143	0.050	0.975	0.867	2
threshold:3m	conditional_connectivity_pct	68	0.054	0.043	0.662	0.704	0.556	8
threshold:3m	connected_area_km2	68	-0.292	-0.179	0.016	0.936	0.750	4
neighbours:8	connected_reference_land_pct	68	-0.320	-0.201	0.008	0.961	0.647	6
neighbours:8	below_reference_land_pct	68	-0.309	-0.188	0.011	1.000	1.000	0
neighbours:8	conditional_connectivity_pct	68	-0.006	-4.434e-04	0.964	0.901	0.867	2
neighbours:8	connected_area_km2	68	-0.329	-0.203	0.005	0.967	0.647	6
seed:local-ocean-only	connected_reference_land_pct	65	-0.394	-0.258	0.002	0.781	0.733	4
seed:local-ocean-only	below_reference_land_pct	65	-0.294	-0.184	0.019	1.000	1.000	0
seed:local-ocean-only	conditional_connectivity_pct	65	-0.061	-0.048	0.631	0.865	0.733	4
seed:local-ocean-only	connected_area_km2	65	-0.423	-0.272	4.000e-04	0.780	0.733	4
uniform-elevation-offset:-1m	connected_reference_land_pct	68	-0.295	-0.177	0.015	0.938	0.750	4

Setting	Terrain metric	n	Spearman rho	Kendall tau	Two-sided p	Terrain-rank rho vs primary	Top-20% Jaccard vs primary	Top-20% membership changes
uniform-elevation-offset:-1m	below_reference_land_pct	68	-0.238	-0.143	0.053	0.975	0.867	2
uniform-elevation-offset:-1m	conditional_connectivity_pct	68	0.054	0.043	0.649	0.704	0.556	8
uniform-elevation-offset:-1m	connected_area_km2	68	-0.292	-0.179	0.015	0.936	0.750	4
uniform-elevation-offset:-0.5m	connected_reference_land_pct	68	-0.310	-0.191	0.010	0.949	0.750	4
uniform-elevation-offset:-0.5m	below_reference_land_pct	68	-0.277	-0.169	0.022	0.990	1.000	0
uniform-elevation-offset:-0.5m	conditional_connectivity_pct	68	0.066	0.047	0.592	0.804	0.867	2
uniform-elevation-offset:-0.5m	connected_area_km2	68	-0.308	-0.188	0.010	0.956	0.750	4
uniform-elevation-offset:+0.5m	connected_reference_land_pct	68	-0.394	-0.260	7.999e-04	0.938	0.750	4
uniform-elevation-offset:+0.5m	below_reference_land_pct	68	-0.320	-0.200	0.008	0.993	1.000	0
uniform-elevation-offset:+0.5m	conditional_connectivity_pct	68	-0.015	-0.014	0.903	0.838	0.556	8
uniform-elevation-offset:+0.5m	connected_area_km2	68	-0.396	-0.250	0.001	0.940	0.867	2
uniform-elevation-offset:+1m	connected_reference_land_pct	68	-0.418	-0.271	2.000e-04	0.844	0.750	4
uniform-elevation-offset:+1m	below_reference_land_pct	68	-0.360	-0.239	0.002	0.950	0.867	2
uniform-elevation-offset:+1m	conditional_connectivity_pct	68	0.045	0.029	0.712	0.783	0.647	6
uniform-elevation-offset:+1m	connected_area_km2	68	-0.422	-0.273	4.000e-04	0.844	0.750	4
COAST-match:<=1km	connected_reference_land_pct	56	-0.420	-0.273	0.001	1.000	1.000	0
COAST-match:<=1km	below_reference_land_pct	56	-0.376	-0.232	0.004	1.000	1.000	0

Setting	Terrain metric	n	Spearman rho	Kendall tau	Two-sided p	Terrain-rank rho vs primary	Top-20% Jaccard vs primary	Top-20% membership changes
COAST-match:<=1km	conditional_connectivity_pct	56	0.146	0.095	0.286	1.000	1.000	0
COAST-match:<=1km	connected_area_km2	56	-0.414	-0.264	0.002	1.000	1.000	0
COAST-match:<=2km	connected_reference_land_pct	65	-0.334	-0.210	0.005	1.000	1.000	0
COAST-match:<=2km	below_reference_land_pct	65	-0.315	-0.191	0.012	1.000	1.000	0
COAST-match:<=2km	conditional_connectivity_pct	65	0.162	0.108	0.205	1.000	1.000	0
COAST-match:<=2km	connected_area_km2	65	-0.335	-0.202	0.007	1.000	1.000	0
COAST-match:distance-unrestricted	connected_reference_land_pct	74	-0.311	-0.196	0.007	1.000	1.000	0
COAST-match:distance-unrestricted	below_reference_land_pct	74	-0.279	-0.163	0.015	1.000	1.000	0
COAST-match:distance-unrestricted	conditional_connectivity_pct	74	0.113	0.079	0.337	1.000	1.000	0
COAST-match:distance-unrestricted	connected_area_km2	74	-0.324	-0.198	0.006	1.000	1.000	0

Supplementary Table S9. Spatially constrained top-k overlap null curves, with nonspatial hypergeometric comparators.

Top-set share (%)	n	Top-set size	Block scale (km)	Spatial-block field	Number of blocks	Observed overlap	Within-block null median	Within-block null 2.5%	Within-block null 97.5%	Within-block lower-tail p	Nonspatial hypergeometric mean	Nonspatial hypergeometric 2.5%	Nonspatial hypergeometric 97.5%
10	68	7	100	spatial_block_10okm	42	0	0.000	0.000	0.000	1.000	0.721	0.000	2.000
10	68	7	150	spatial_block_15okm	34	0	0.000	0.000	0.000	1.000	0.721	0.000	2.000
10	68	7	250	spatial_block_25okm	25	0	0.000	0.000	0.000	1.000	0.721	0.000	2.000
15	68	11	100	spatial_block_10okm	42	0	0.000	0.000	0.000	1.000	1.779	0.000	4.000
15	68	11	150	spatial_block_15okm	34	0	1.000	0.000	1.000	0.327	1.779	0.000	4.000

Top-set share (%)	n	Top-set size	Block scale (km)	Spatial-block field	Number of blocks	Observed overlap	Within-block null median	Within-block null 2.5%	Within-block null 97.5%	Within-block lower-tail p	Nonspatial hypergeometric mean	Nonspatial hypergeometric 2.5%	Nonspatial hypergeometric 97.5%
15	68	11	250	spatial_block_250km	25	0	1.000	0.000	1.000	0.334	1.779	0.000	4.000
20	68	14	100	spatial_block_100km	42	1	1.000	1.000	1.000	1.000	2.882	0.000	6.000
20	68	14	150	spatial_block_150km	34	1	2.000	1.000	2.000	0.327	2.882	0.000	6.000
20	68	14	250	spatial_block_250km	25	1	3.000	1.000	4.000	0.098	2.882	0.000	6.000
25	68	17	100	spatial_block_100km	42	2	3.000	2.000	3.000	0.336	4.250	1.000	7.000
25	68	17	150	spatial_block_150km	34	2	3.000	2.000	4.000	0.223	4.250	1.000	7.000
25	68	17	250	spatial_block_250km	25	2	3.000	2.000	5.000	0.097	4.250	1.000	7.000
30	68	21	100	spatial_block_100km	42	5	6.000	5.000	6.000	0.336	6.485	3.000	10.000
30	68	21	150	spatial_block_150km	34	5	6.000	5.000	7.000	0.164	6.485	3.000	10.000
30	68	21	250	spatial_block_250km	25	5	7.000	5.000	8.000	0.072	6.485	3.000	10.000
35	68	24	100	spatial_block_100km	42	5	6.000	5.000	6.000	0.336	8.471	5.000	12.000
35	68	24	150	spatial_block_150km	34	5	7.000	5.000	8.000	0.041	8.471	5.000	12.000
35	68	24	250	spatial_block_250km	25	5	7.000	6.000	9.000	0.015	8.471	5.000	12.000
40	68	28	100	spatial_block_100km	42	8	9.000	8.000	9.000	0.336	11.529	8.000	15.000
40	68	28	150	spatial_block_150km	34	8	10.000	8.000	10.000	0.082	11.529	8.000	15.000
40	68	28	250	spatial_block_250km	25	8	10.000	8.000	11.000	0.028	11.529	8.000	15.000

The table reports five-percentage-point intervals for readability. The machine-readable source file retains every integer top-set share from 10% to 40%. Hypergeometric values are descriptive nonspatial comparators; inference uses within-block permutations at each target scale.

Supplementary Table S10. Leave-one-spatial-block-out calculations at three scales.

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_reference_land_pct	100	B100_DE01	1	67	-0.360	-0.230	14	1
connected_reference_land_pct	100	B100_DK01	1	67	-0.343	-0.219	14	1
connected_reference_land_pct	100	B100_FRB01	1	67	-0.335	-0.215	14	1
connected_reference_land_pct	100	B100_FRB02	2	66	-0.333	-0.207	14	1
connected_reference_land_pct	100	B100_FRB03	2	66	-0.320	-0.201	14	1
connected_reference_land_pct	100	B100_FRC01	2	66	-0.306	-0.195	14	1
connected_reference_land_pct	100	B100_FRC02	2	66	-0.313	-0.195	14	1
connected_reference_land_pct	100	B100_FRC03	2	66	-0.329	-0.207	14	1
connected_reference_land_pct	100	B100_FRC04	1	67	-0.328	-0.209	14	1
connected_reference_land_pct	100	B100_FRC05	2	66	-0.387	-0.250	14	1
connected_reference_land_pct	100	B100_FRL01	2	66	-0.338	-0.214	14	1
connected_reference_land_pct	100	B100_NL01	1	67	-0.341	-0.218	14	1
connected_reference_land_pct	100	B100_NL02	1	67	-0.315	-0.197	14	1
connected_reference_land_pct	100	B100_NL03	1	67	-0.353	-0.225	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_reference_land_pct	100	B100_NO01	1	67	-0.320	-0.198	14	1
connected_reference_land_pct	100	B100_UKBC01	1	67	-0.308	-0.194	14	1
connected_reference_land_pct	100	B100_UKBC02	4	64	-0.233	-0.147	13	2
connected_reference_land_pct	100	B100_UKBC03	1	67	-0.310	-0.195	14	1
connected_reference_land_pct	100	B100_UKC01	2	66	-0.340	-0.216	14	1
connected_reference_land_pct	100	B100_UKC02	1	67	-0.325	-0.204	14	1
connected_reference_land_pct	100	B100_UKC03	1	67	-0.346	-0.218	14	1
connected_reference_land_pct	100	B100_UKC04	1	67	-0.321	-0.200	14	1
connected_reference_land_pct	100	B100_UKI01	2	66	-0.349	-0.221	14	1
connected_reference_land_pct	100	B100_UKI02	3	65	-0.337	-0.218	13	0
connected_reference_land_pct	100	B100_UKI03	3	65	-0.303	-0.191	13	1
connected_reference_land_pct	100	B100_UKI04	2	66	-0.341	-0.217	14	1
connected_reference_land_pct	100	B100_UKI05	2	66	-0.337	-0.212	14	1
connected_reference_land_pct	100	B100_UKNE01	2	66	-0.332	-0.209	14	1
connected_reference_land_pct	100	B100_UKNE02	1	67	-0.323	-0.203	14	1
connected_reference_land_pct	100	B100_UKNE03	1	67	-0.336	-0.211	14	1
connected_reference_land_pct	100	B100_UKNE04	2	66	-0.353	-0.222	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_reference_land_pct	100	B100_UKNE05	2	66	-0.329	-0.207	14	1
connected_reference_land_pct	100	B100_UKNE06	1	67	-0.332	-0.211	14	1
connected_reference_land_pct	100	B100_UKNE07	1	67	-0.329	-0.206	14	1
connected_reference_land_pct	100	B100_UKSE01	1	67	-0.364	-0.233	14	0
connected_reference_land_pct	100	B100_UKSE02	2	66	-0.369	-0.242	14	1
connected_reference_land_pct	100	B100_UKSW01	2	66	-0.344	-0.216	14	1
connected_reference_land_pct	100	B100_UKSW02	1	67	-0.335	-0.209	14	1
connected_reference_land_pct	100	B100_UKWA01	3	65	-0.345	-0.226	13	0
connected_reference_land_pct	100	B100_UKWA02	1	67	-0.336	-0.209	14	1
connected_reference_land_pct	100	B100_UKWA03	2	66	-0.346	-0.214	14	1
connected_reference_land_pct	100	B100_UKWA04	1	67	-0.332	-0.211	14	1
below_reference_land_pct	100	B100_DE01	1	67	-0.335	-0.207	14	1
below_reference_land_pct	100	B100_DK01	1	67	-0.318	-0.194	14	1
below_reference_land_pct	100	B100_FRB01	1	67	-0.312	-0.192	14	1
below_reference_land_pct	100	B100_FRB02	2	66	-0.311	-0.186	14	1
below_reference_land_pct	100	B100_FRB03	2	66	-0.294	-0.178	14	1
below_reference_land_pct	100	B100_FRC01	2	66	-0.279	-0.168	14	1
below_reference_land_pct	100	B100_FRC02	2	66	-0.287	-0.170	14	1
below_reference_land_pct	100	B100_FRC03	2	66	-0.296	-0.177	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
below_reference_land_pct	100	B100_FRC04	1	67	-0.297	-0.181	14	1
below_reference_land_pct	100	B100_FRC05	2	66	-0.358	-0.225	14	1
below_reference_land_pct	100	B100_FRL01	2	66	-0.318	-0.198	14	1
below_reference_land_pct	100	B100_NL01	1	67	-0.316	-0.192	14	1
below_reference_land_pct	100	B100_NL02	1	67	-0.290	-0.173	14	1
below_reference_land_pct	100	B100_NL03	1	67	-0.329	-0.201	14	1
below_reference_land_pct	100	B100_NO01	1	67	-0.301	-0.179	14	1
below_reference_land_pct	100	B100_UKBC01	1	67	-0.282	-0.170	14	1
below_reference_land_pct	100	B100_UKBC02	4	64	-0.200	-0.114	13	2
below_reference_land_pct	100	B100_UKBC03	1	67	-0.283	-0.171	14	1
below_reference_land_pct	100	B100_UKC01	2	66	-0.316	-0.193	14	1
below_reference_land_pct	100	B100_UKC02	1	67	-0.302	-0.182	14	1
below_reference_land_pct	100	B100_UKC03	1	67	-0.321	-0.195	14	1
below_reference_land_pct	100	B100_UKC04	1	67	-0.297	-0.179	14	1
below_reference_land_pct	100	B100_UKI01	2	66	-0.325	-0.199	14	1
below_reference_land_pct	100	B100_UKI02	3	65	-0.314	-0.193	13	0
below_reference_land_pct	100	B100_UKI03	3	65	-0.269	-0.161	13	1
below_reference_land_pct	100	B100_UKI04	2	66	-0.315	-0.192	14	1
below_reference_land_pct	100	B100_UKI05	2	66	-0.318	-0.193	14	1
below_reference_land_pct	100	B100_UKNE01	2	66	-0.299	-0.178	14	1
below_reference_land_pct	100	B100_UKNE02	1	67	-0.331	-0.204	14	1
below_reference_land_pct	100	B100_UKNE03	1	67	-0.310	-0.186	14	1
below_reference_land_pct	100	B100_UKNE04	2	66	-0.327	-0.197	14	1
below_reference_land_pct	100	B100_UKNE05	2	66	-0.306	-0.187	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
below_reference_land_pct	100	B100_UKNE06	1	67	-0.310	-0.190	14	1
below_reference_land_pct	100	B100_UKNE07	1	67	-0.308	-0.186	14	1
below_reference_land_pct	100	B100_UKSE01	1	67	-0.337	-0.209	14	0
below_reference_land_pct	100	B100_UKSE02	2	66	-0.344	-0.216	14	1
below_reference_land_pct	100	B100_UKSW01	2	66	-0.325	-0.198	14	1
below_reference_land_pct	100	B100_UKSW02	1	67	-0.311	-0.187	14	1
below_reference_land_pct	100	B100_UKWA01	3	65	-0.324	-0.206	13	0
below_reference_land_pct	100	B100_UKWA02	1	67	-0.315	-0.189	14	1
below_reference_land_pct	100	B100_UKWA03	2	66	-0.330	-0.197	14	1
below_reference_land_pct	100	B100_UKWA04	1	67	-0.308	-0.189	14	1
conditional_connectivity_pct	100	B100_DE01	1	67	0.097	0.063	14	5
conditional_connectivity_pct	100	B100_DK01	1	67	0.104	0.069	14	5
conditional_connectivity_pct	100	B100_FRB01	1	67	0.096	0.063	14	5
conditional_connectivity_pct	100	B100_FRB02	2	66	0.133	0.089	14	5
conditional_connectivity_pct	100	B100_FRB03	2	66	0.117	0.078	14	5
conditional_connectivity_pct	100	B100_FRC01	2	66	0.129	0.083	14	5
conditional_connectivity_pct	100	B100_FRC02	2	66	0.072	0.048	14	4
conditional_connectivity_pct	100	B100_FRC03	2	66	0.113	0.073	14	5
conditional_connectivity_pct	100	B100_FRC04	1	67	0.076	0.051	14	4
conditional_connectivity_pct	100	B100_FRC05	2	66	0.120	0.080	14	5
conditional_connectivity_pct	100	B100_FRL01	2	66	0.112	0.073	14	5
conditional_connectivity_pct	100	B100_NL01	1	67	0.106	0.069	14	5
conditional_connectivity_pct	100	B100_NL02	1	67	0.123	0.081	14	5
conditional_connectivity_pct	100	B100_NL03	1	67	0.096	0.062	14	5

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
conditional_connectivity_pct	100	B100_NO01	1	67	0.093	0.063	14	5
conditional_connectivity_pct	100	B100_UKBC01	1	67	0.075	0.050	14	4
conditional_connectivity_pct	100	B100_UKBC02	4	64	0.047	0.028	13	4
conditional_connectivity_pct	100	B100_UKBC03	1	67	0.101	0.066	14	5
conditional_connectivity_pct	100	B100_UKC01	2	66	0.055	0.036	14	5
conditional_connectivity_pct	100	B100_UKC02	1	67	0.100	0.066	14	5
conditional_connectivity_pct	100	B100_UKC03	1	67	0.117	0.077	14	5
conditional_connectivity_pct	100	B100_UKC04	1	67	0.101	0.066	14	5
conditional_connectivity_pct	100	B100_UKI01	2	66	0.107	0.073	14	5
conditional_connectivity_pct	100	B100_UKI02	3	65	0.124	0.084	13	5
conditional_connectivity_pct	100	B100_UKI03	3	65	0.077	0.052	13	4
conditional_connectivity_pct	100	B100_UKI04	2	66	0.134	0.089	14	6
conditional_connectivity_pct	100	B100_UKI05	2	66	0.115	0.078	14	5
conditional_connectivity_pct	100	B100_UKNE01	2	66	0.069	0.046	14	5
conditional_connectivity_pct	100	B100_UKNE02	1	67	0.123	0.083	14	5
conditional_connectivity_pct	100	B100_UKNE03	1	67	0.105	0.068	14	5
conditional_connectivity_pct	100	B100_UKNE04	2	66	0.106	0.071	14	5
conditional_connectivity_pct	100	B100_UKNE05	2	66	0.098	0.065	14	5
conditional_connectivity_pct	100	B100_UKNE06	1	67	0.122	0.080	14	5
conditional_connectivity_pct	100	B100_UKNE07	1	67	0.110	0.073	14	5
conditional_connectivity_pct	100	B100_UKSE01	1	67	0.101	0.065	14	5
conditional_connectivity_pct	100	B100_UKSE02	2	66	0.106	0.071	14	5
conditional_connectivity_pct	100	B100_UKSW01	2	66	0.115	0.080	14	5
conditional_connectivity_pct	100	B100_UKSW02	1	67	0.112	0.075	14	5

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
conditional_connectivity_pct	100	B100_UKWA01	3	65	0.141	0.093	13	5
conditional_connectivity_pct	100	B100_UKWA02	1	67	0.118	0.078	14	5
conditional_connectivity_pct	100	B100_UKWA03	2	66	0.122	0.082	14	5
conditional_connectivity_pct	100	B100_UKWA04	1	67	0.097	0.065	14	5
connected_area_km2	100	B100_DE01	1	67	-0.370	-0.228	14	1
connected_area_km2	100	B100_DK01	1	67	-0.351	-0.218	14	1
connected_area_km2	100	B100_FRB01	1	67	-0.342	-0.208	14	1
connected_area_km2	100	B100_FRB02	2	66	-0.352	-0.213	14	1
connected_area_km2	100	B100_FRB03	2	66	-0.328	-0.199	14	1
connected_area_km2	100	B100_FRC01	2	66	-0.306	-0.187	14	1
connected_area_km2	100	B100_FRC02	2	66	-0.327	-0.200	14	1
connected_area_km2	100	B100_FRC03	2	66	-0.339	-0.206	14	1
connected_area_km2	100	B100_FRC04	1	67	-0.338	-0.206	14	1
connected_area_km2	100	B100_FRC05	2	66	-0.394	-0.248	14	1
connected_area_km2	100	B100_FRL01	2	66	-0.345	-0.209	14	1
connected_area_km2	100	B100_NL01	1	67	-0.349	-0.216	14	1
connected_area_km2	100	B100_NL02	1	67	-0.324	-0.195	14	1
connected_area_km2	100	B100_NL03	1	67	-0.361	-0.222	14	1
connected_area_km2	100	B100_NO01	1	67	-0.329	-0.195	14	1
connected_area_km2	100	B100_UKBC01	1	67	-0.317	-0.191	14	1
connected_area_km2	100	B100_UKBC02	4	64	-0.248	-0.148	13	2
connected_area_km2	100	B100_UKBC03	1	67	-0.315	-0.190	14	1
connected_area_km2	100	B100_UKC01	2	66	-0.342	-0.210	14	1
connected_area_km2	100	B100_UKC02	1	67	-0.332	-0.200	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_area_km2	100	B100_UKC03	1	67	-0.356	-0.218	14	1
connected_area_km2	100	B100_UKC04	1	67	-0.330	-0.200	14	1
connected_area_km2	100	B100_UKI01	2	66	-0.362	-0.221	14	1
connected_area_km2	100	B100_UKI02	3	65	-0.343	-0.214	13	0
connected_area_km2	100	B100_UKI03	3	65	-0.300	-0.184	13	1
connected_area_km2	100	B100_UKI04	2	66	-0.345	-0.211	14	1
connected_area_km2	100	B100_UKI05	2	66	-0.347	-0.210	14	1
connected_area_km2	100	B100_UKNE01	2	66	-0.339	-0.205	14	1
connected_area_km2	100	B100_UKNE02	1	67	-0.332	-0.200	14	1
connected_area_km2	100	B100_UKNE03	1	67	-0.344	-0.208	14	1
connected_area_km2	100	B100_UKNE04	2	66	-0.362	-0.221	14	1
connected_area_km2	100	B100_UKNE05	2	66	-0.335	-0.204	14	1
connected_area_km2	100	B100_UKNE06	1	67	-0.338	-0.206	14	1
connected_area_km2	100	B100_UKNE07	1	67	-0.337	-0.203	14	1
connected_area_km2	100	B100_UKSE01	1	67	-0.372	-0.230	14	0
connected_area_km2	100	B100_UKSE02	2	66	-0.378	-0.240	14	1
connected_area_km2	100	B100_UKSW01	2	66	-0.352	-0.215	14	1
connected_area_km2	100	B100_UKSW02	1	67	-0.341	-0.207	14	1
connected_area_km2	100	B100_UKWA01	3	65	-0.351	-0.223	13	0
connected_area_km2	100	B100_UKWA02	1	67	-0.344	-0.209	14	1
connected_area_km2	100	B100_UKWA03	2	66	-0.346	-0.207	14	1
connected_area_km2	100	B100_UKWA04	1	67	-0.339	-0.208	14	1
connected_reference_land_pct	150	B150_DE01	2	66	-0.381	-0.246	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_reference_land_pct	150	B150_DK01	1	67	-0.343	-0.219	14	1
connected_reference_land_pct	150	B150_FRB01	3	65	-0.334	-0.210	13	0
connected_reference_land_pct	150	B150_FRB02	2	66	-0.320	-0.201	14	1
connected_reference_land_pct	150	B150_FRC01	2	66	-0.306	-0.195	14	1
connected_reference_land_pct	150	B150_FRC02	2	66	-0.313	-0.195	14	1
connected_reference_land_pct	150	B150_FRC03	2	66	-0.329	-0.207	14	1
connected_reference_land_pct	150	B150_FRC04	3	65	-0.384	-0.250	13	1
connected_reference_land_pct	150	B150_FRL01	2	66	-0.338	-0.214	14	1
connected_reference_land_pct	150	B150_NL01	1	67	-0.341	-0.218	14	1
connected_reference_land_pct	150	B150_NL02	1	67	-0.315	-0.197	14	1
connected_reference_land_pct	150	B150_NO01	1	67	-0.320	-0.198	14	1
connected_reference_land_pct	150	B150_UKBC01	1	67	-0.308	-0.194	14	1
connected_reference_land_pct	150	B150_UKBC02	4	64	-0.233	-0.147	13	2
connected_reference_land_pct	150	B150_UKBC03	1	67	-0.310	-0.195	14	1
connected_reference_land_pct	150	B150_UKC01	2	66	-0.340	-0.216	14	1
connected_reference_land_pct	150	B150_UKC02	1	67	-0.325	-0.204	14	1
connected_reference_land_pct	150	B150_UKC03	2	66	-0.335	-0.208	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_reference_land_pct	150	B150_UKI01	2	66	-0.349	-0.221	14	1
connected_reference_land_pct	150	B150_UKI02	3	65	-0.337	-0.218	13	0
connected_reference_land_pct	150	B150_UKI03	4	64	-0.314	-0.198	13	1
connected_reference_land_pct	150	B150_UKI04	3	65	-0.336	-0.213	13	0
connected_reference_land_pct	150	B150_UKNE01	2	66	-0.332	-0.209	14	1
connected_reference_land_pct	150	B150_UKNE02	1	67	-0.323	-0.203	14	1
connected_reference_land_pct	150	B150_UKNE03	1	67	-0.336	-0.211	14	1
connected_reference_land_pct	150	B150_UKNE04	2	66	-0.353	-0.222	14	1
connected_reference_land_pct	150	B150_UKNE05	3	65	-0.327	-0.208	13	0
connected_reference_land_pct	150	B150_UKNE06	1	67	-0.329	-0.206	14	1
connected_reference_land_pct	150	B150_UKSE01	1	67	-0.364	-0.233	14	0
connected_reference_land_pct	150	B150_UKSE02	2	66	-0.369	-0.242	14	1
connected_reference_land_pct	150	B150_UKSW01	3	65	-0.345	-0.213	13	0
connected_reference_land_pct	150	B150_UKWA01	3	65	-0.345	-0.226	13	0
connected_reference_land_pct	150	B150_UKWA02	1	67	-0.336	-0.209	14	1
connected_reference_land_pct	150	B150_UKWA03	3	65	-0.346	-0.215	13	0
below_reference_land_pct	150	B150_DE01	2	66	-0.356	-0.221	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
below_reference_land_pct	150	B150_DK01	1	67	-0.318	-0.194	14	1
below_reference_land_pct	150	B150_FRB01	3	65	-0.312	-0.189	13	0
below_reference_land_pct	150	B150_FRB02	2	66	-0.294	-0.178	14	1
below_reference_land_pct	150	B150_FRC01	2	66	-0.279	-0.168	14	1
below_reference_land_pct	150	B150_FRC02	2	66	-0.287	-0.170	14	1
below_reference_land_pct	150	B150_FRC03	2	66	-0.296	-0.177	14	1
below_reference_land_pct	150	B150_FRC04	3	65	-0.348	-0.220	13	1
below_reference_land_pct	150	B150_FRL01	2	66	-0.318	-0.198	14	1
below_reference_land_pct	150	B150_NL01	1	67	-0.316	-0.192	14	1
below_reference_land_pct	150	B150_NL02	1	67	-0.290	-0.173	14	1
below_reference_land_pct	150	B150_NO01	1	67	-0.301	-0.179	14	1
below_reference_land_pct	150	B150_UKBC01	1	67	-0.282	-0.170	14	1
below_reference_land_pct	150	B150_UKBC02	4	64	-0.200	-0.114	13	2
below_reference_land_pct	150	B150_UKBC03	1	67	-0.283	-0.171	14	1
below_reference_land_pct	150	B150_UKC01	2	66	-0.316	-0.193	14	1
below_reference_land_pct	150	B150_UKC02	1	67	-0.302	-0.182	14	1
below_reference_land_pct	150	B150_UKC03	2	66	-0.309	-0.186	14	1
below_reference_land_pct	150	B150_UKI01	2	66	-0.325	-0.199	14	1
below_reference_land_pct	150	B150_UKI02	3	65	-0.314	-0.193	13	0
below_reference_land_pct	150	B150_UKI03	4	64	-0.275	-0.164	13	1
below_reference_land_pct	150	B150_UKI04	3	65	-0.319	-0.195	13	0
below_reference_land_pct	150	B150_UKNE01	2	66	-0.299	-0.178	14	1
below_reference_land_pct	150	B150_UKNE02	1	67	-0.331	-0.204	14	1
below_reference_land_pct	150	B150_UKNE03	1	67	-0.310	-0.186	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
below_reference_land_pct	150	B150_UKNE04	2	66	-0.327	-0.197	14	1
below_reference_land_pct	150	B150_UKNE05	3	65	-0.306	-0.188	13	0
below_reference_land_pct	150	B150_UKNE06	1	67	-0.308	-0.186	14	1
below_reference_land_pct	150	B150_UKSE01	1	67	-0.337	-0.209	14	0
below_reference_land_pct	150	B150_UKSE02	2	66	-0.344	-0.216	14	1
below_reference_land_pct	150	B150_UKSW01	3	65	-0.327	-0.197	13	0
below_reference_land_pct	150	B150_UKWA01	3	65	-0.324	-0.206	13	0
below_reference_land_pct	150	B150_UKWA02	1	67	-0.315	-0.189	14	1
below_reference_land_pct	150	B150_UKWA03	3	65	-0.330	-0.199	13	0
conditional_connectivity_pct	150	B150_DE01	2	66	0.088	0.054	14	5
conditional_connectivity_pct	150	B150_DK01	1	67	0.104	0.069	14	5
conditional_connectivity_pct	150	B150_FRB01	3	65	0.122	0.082	13	5
conditional_connectivity_pct	150	B150_FRB02	2	66	0.117	0.078	14	5
conditional_connectivity_pct	150	B150_FRC01	2	66	0.129	0.083	14	5
conditional_connectivity_pct	150	B150_FRC02	2	66	0.072	0.048	14	4
conditional_connectivity_pct	150	B150_FRC03	2	66	0.113	0.073	14	5
conditional_connectivity_pct	150	B150_FRC04	3	65	0.093	0.061	13	4
conditional_connectivity_pct	150	B150_FRL01	2	66	0.112	0.073	14	5
conditional_connectivity_pct	150	B150_NL01	1	67	0.106	0.069	14	5
conditional_connectivity_pct	150	B150_NL02	1	67	0.123	0.081	14	5
conditional_connectivity_pct	150	B150_NO01	1	67	0.093	0.063	14	5
conditional_connectivity_pct	150	B150_UKBC01	1	67	0.075	0.050	14	4
conditional_connectivity_pct	150	B150_UKBC02	4	64	0.047	0.028	13	4
conditional_connectivity_pct	150	B150_UKBC03	1	67	0.101	0.066	14	5

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
conditional_connectivity_pct	150	B150_UKC01	2	66	0.055	0.036	14	5
conditional_connectivity_pct	150	B150_UKC02	1	67	0.100	0.066	14	5
conditional_connectivity_pct	150	B150_UKC03	2	66	0.114	0.075	14	5
conditional_connectivity_pct	150	B150_UKI01	2	66	0.107	0.073	14	5
conditional_connectivity_pct	150	B150_UKI02	3	65	0.124	0.084	13	5
conditional_connectivity_pct	150	B150_UKI03	4	64	0.071	0.048	13	4
conditional_connectivity_pct	150	B150_UKI04	3	65	0.147	0.100	13	5
conditional_connectivity_pct	150	B150_UKNE01	2	66	0.069	0.046	14	5
conditional_connectivity_pct	150	B150_UKNE02	1	67	0.123	0.083	14	5
conditional_connectivity_pct	150	B150_UKNE03	1	67	0.105	0.068	14	5
conditional_connectivity_pct	150	B150_UKNE04	2	66	0.106	0.071	14	5
conditional_connectivity_pct	150	B150_UKNE05	3	65	0.115	0.075	13	5
conditional_connectivity_pct	150	B150_UKNE06	1	67	0.110	0.073	14	5
conditional_connectivity_pct	150	B150_UKSE01	1	67	0.101	0.065	14	5
conditional_connectivity_pct	150	B150_UKSE02	2	66	0.106	0.071	14	5
conditional_connectivity_pct	150	B150_UKSW01	3	65	0.124	0.086	13	5
conditional_connectivity_pct	150	B150_UKWA01	3	65	0.141	0.093	13	5
conditional_connectivity_pct	150	B150_UKWA02	1	67	0.118	0.078	14	5
conditional_connectivity_pct	150	B150_UKWA03	3	65	0.116	0.079	13	5
connected_area_km2	150	B150_DE01	2	66	-0.391	-0.244	14	1
connected_area_km2	150	B150_DK01	1	67	-0.351	-0.218	14	1
connected_area_km2	150	B150_FRB01	3	65	-0.353	-0.212	13	0
connected_area_km2	150	B150_FRB02	2	66	-0.328	-0.199	14	1
connected_area_km2	150	B150_FRC01	2	66	-0.306	-0.187	14	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_area_km2	150	B150_FRC02	2	66	-0.327	-0.200	14	1
connected_area_km2	150	B150_FRC03	2	66	-0.339	-0.206	14	1
connected_area_km2	150	B150_FRC04	3	65	-0.394	-0.247	13	1
connected_area_km2	150	B150_FRL01	2	66	-0.345	-0.209	14	1
connected_area_km2	150	B150_NL01	1	67	-0.349	-0.216	14	1
connected_area_km2	150	B150_NL02	1	67	-0.324	-0.195	14	1
connected_area_km2	150	B150_NO01	1	67	-0.329	-0.195	14	1
connected_area_km2	150	B150_UKBC01	1	67	-0.317	-0.191	14	1
connected_area_km2	150	B150_UKBC02	4	64	-0.248	-0.148	13	2
connected_area_km2	150	B150_UKBC03	1	67	-0.315	-0.190	14	1
connected_area_km2	150	B150_UKC01	2	66	-0.342	-0.210	14	1
connected_area_km2	150	B150_UKC02	1	67	-0.332	-0.200	14	1
connected_area_km2	150	B150_UKC03	2	66	-0.346	-0.210	14	1
connected_area_km2	150	B150_UKI01	2	66	-0.362	-0.221	14	1
connected_area_km2	150	B150_UKI02	3	65	-0.343	-0.214	13	0
connected_area_km2	150	B150_UKI03	4	64	-0.307	-0.189	13	1
connected_area_km2	150	B150_UKI04	3	65	-0.344	-0.209	13	0
connected_area_km2	150	B150_UKNE01	2	66	-0.339	-0.205	14	1
connected_area_km2	150	B150_UKNE02	1	67	-0.332	-0.200	14	1
connected_area_km2	150	B150_UKNE03	1	67	-0.344	-0.208	14	1
connected_area_km2	150	B150_UKNE04	2	66	-0.362	-0.221	14	1
connected_area_km2	150	B150_UKNE05	3	65	-0.331	-0.201	13	0
connected_area_km2	150	B150_UKNE06	1	67	-0.337	-0.203	14	1
connected_area_km2	150	B150_UKSE01	1	67	-0.372	-0.230	14	0

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_area_km2	150	B150_UKSE02	2	66	-0.378	-0.240	14	1
connected_area_km2	150	B150_UKSW01	3	65	-0.352	-0.213	13	0
connected_area_km2	150	B150_UKWA01	3	65	-0.351	-0.223	13	0
connected_area_km2	150	B150_UKWA02	1	67	-0.344	-0.209	14	1
connected_area_km2	150	B150_UKWA03	3	65	-0.345	-0.208	13	0
connected_reference_land_pct	250	B250_DE01	2	66	-0.381	-0.246	14	1
connected_reference_land_pct	250	B250_DK01	1	67	-0.343	-0.219	14	1
connected_reference_land_pct	250	B250_FRB01	3	65	-0.334	-0.210	13	0
connected_reference_land_pct	250	B250_FRB02	3	65	-0.309	-0.193	13	0
connected_reference_land_pct	250	B250_FRC01	3	65	-0.293	-0.185	13	1
connected_reference_land_pct	250	B250_FRC02	2	66	-0.329	-0.207	14	1
connected_reference_land_pct	250	B250_FRC03	3	65	-0.384	-0.250	13	1
connected_reference_land_pct	250	B250_FRL01	2	66	-0.338	-0.214	14	1
connected_reference_land_pct	250	B250_NL01	2	66	-0.322	-0.204	14	1
connected_reference_land_pct	250	B250_NO01	1	67	-0.320	-0.198	14	1
connected_reference_land_pct	250	B250_UKBC01	5	63	-0.201	-0.125	13	3
connected_reference_land_pct	250	B250_UKBC02	1	67	-0.310	-0.195	14	1
connected_reference_land_pct	250	B250_UKC01	3	65	-0.330	-0.208	13	0

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_reference_land_pct	250	B250_UKC02	2	66	-0.335	-0.208	14	1
connected_reference_land_pct	250	B250_UKI01	2	66	-0.349	-0.221	14	1
connected_reference_land_pct	250	B250_UKI02	7	61	-0.320	-0.209	13	2
connected_reference_land_pct	250	B250_UKI03	3	65	-0.336	-0.213	13	0
connected_reference_land_pct	250	B250_UKNE01	2	66	-0.332	-0.209	14	1
connected_reference_land_pct	250	B250_UKNE02	1	67	-0.323	-0.203	14	1
connected_reference_land_pct	250	B250_UKNE03	3	65	-0.357	-0.223	13	0
connected_reference_land_pct	250	B250_UKNE04	4	64	-0.322	-0.202	13	0
connected_reference_land_pct	250	B250_UKSE01	3	65	-0.402	-0.265	13	0
connected_reference_land_pct	250	B250_UKSW01	3	65	-0.345	-0.213	13	0
connected_reference_land_pct	250	B250_UKWA01	4	64	-0.349	-0.226	13	0
connected_reference_land_pct	250	B250_UKWA02	3	65	-0.346	-0.215	13	0
below_reference_land_pct	250	B250_DE01	2	66	-0.356	-0.221	14	1
below_reference_land_pct	250	B250_DK01	1	67	-0.318	-0.194	14	1
below_reference_land_pct	250	B250_FRB01	3	65	-0.312	-0.189	13	0
below_reference_land_pct	250	B250_FRB02	3	65	-0.281	-0.167	13	0
below_reference_land_pct	250	B250_FRC01	3	65	-0.267	-0.158	13	1
below_reference_land_pct	250	B250_FRC02	2	66	-0.296	-0.177	14	1
below_reference_land_pct	250	B250_FRC03	3	65	-0.348	-0.220	13	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
below_reference_land_pct	250	B250_FRL01	2	66	-0.318	-0.198	14	1
below_reference_land_pct	250	B250_NL01	2	66	-0.297	-0.178	14	1
below_reference_land_pct	250	B250_NO01	1	67	-0.301	-0.179	14	1
below_reference_land_pct	250	B250_UKBC01	5	63	-0.164	-0.091	13	3
below_reference_land_pct	250	B250_UKBC02	1	67	-0.283	-0.171	14	1
below_reference_land_pct	250	B250_UKC01	3	65	-0.308	-0.187	13	0
below_reference_land_pct	250	B250_UKC02	2	66	-0.309	-0.186	14	1
below_reference_land_pct	250	B250_UKI01	2	66	-0.325	-0.199	14	1
below_reference_land_pct	250	B250_UKI02	7	61	-0.282	-0.172	13	2
below_reference_land_pct	250	B250_UKI03	3	65	-0.319	-0.195	13	0
below_reference_land_pct	250	B250_UKNE01	2	66	-0.299	-0.178	14	1
below_reference_land_pct	250	B250_UKNE02	1	67	-0.331	-0.204	14	1
below_reference_land_pct	250	B250_UKNE03	3	65	-0.328	-0.195	13	0
below_reference_land_pct	250	B250_UKNE04	4	64	-0.305	-0.187	13	0
below_reference_land_pct	250	B250_UKSE01	3	65	-0.374	-0.237	13	0
below_reference_land_pct	250	B250_UKSW01	3	65	-0.327	-0.197	13	0
below_reference_land_pct	250	B250_UKWA01	4	64	-0.332	-0.208	13	0
below_reference_land_pct	250	B250_UKWA02	3	65	-0.330	-0.199	13	0
conditional_connectivity_pct	250	B250_DE01	2	66	0.088	0.054	14	5
conditional_connectivity_pct	250	B250_DK01	1	67	0.104	0.069	14	5
conditional_connectivity_pct	250	B250_FRB01	3	65	0.122	0.082	13	5
conditional_connectivity_pct	250	B250_FRB02	3	65	0.113	0.074	13	5
conditional_connectivity_pct	250	B250_FRC01	3	65	0.100	0.066	13	4
conditional_connectivity_pct	250	B250_FRC02	2	66	0.113	0.073	14	5

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
conditional_connectivity_pct	250	B250_FRC03	3	65	0.093	0.061	13	4
conditional_connectivity_pct	250	B250_FRL01	2	66	0.112	0.073	14	5
conditional_connectivity_pct	250	B250_NL01	2	66	0.124	0.081	14	5
conditional_connectivity_pct	250	B250_NO01	1	67	0.093	0.063	14	5
conditional_connectivity_pct	250	B250_UKBC01	5	63	0.009	0.004	13	3
conditional_connectivity_pct	250	B250_UKBC02	1	67	0.101	0.066	14	5
conditional_connectivity_pct	250	B250_UKC01	3	65	0.050	0.032	13	5
conditional_connectivity_pct	250	B250_UKC02	2	66	0.114	0.075	14	5
conditional_connectivity_pct	250	B250_UKI01	2	66	0.107	0.073	14	5
conditional_connectivity_pct	250	B250_UKI02	7	61	0.095	0.065	13	4
conditional_connectivity_pct	250	B250_UKI03	3	65	0.147	0.100	13	5
conditional_connectivity_pct	250	B250_UKNE01	2	66	0.069	0.046	14	5
conditional_connectivity_pct	250	B250_UKNE02	1	67	0.123	0.083	14	5
conditional_connectivity_pct	250	B250_UKNE03	3	65	0.107	0.070	13	5
conditional_connectivity_pct	250	B250_UKNE04	4	64	0.121	0.078	13	5
conditional_connectivity_pct	250	B250_UKSE01	3	65	0.102	0.067	13	5
conditional_connectivity_pct	250	B250_UKSW01	3	65	0.124	0.086	13	5
conditional_connectivity_pct	250	B250_UKWA01	4	64	0.159	0.105	13	6
conditional_connectivity_pct	250	B250_UKWA02	3	65	0.116	0.079	13	5
connected_area_km2	250	B250_DE01	2	66	-0.391	-0.244	14	1
connected_area_km2	250	B250_DK01	1	67	-0.351	-0.218	14	1
connected_area_km2	250	B250_FRB01	3	65	-0.353	-0.212	13	0
connected_area_km2	250	B250_FRB02	3	65	-0.311	-0.186	13	0
connected_area_km2	250	B250_FRC01	3	65	-0.307	-0.189	13	1

Terrain metric	Block scale (km)	Omitted block	Omitted n	Remaining n	Spearman rho	Kendall tau	Top n	Overlap
connected_area_km2	250	B250_FRC02	2	66	-0.339	-0.206	14	1
connected_area_km2	250	B250_FRC03	3	65	-0.394	-0.247	13	1
connected_area_km2	250	B250_FRL01	2	66	-0.345	-0.209	14	1
connected_area_km2	250	B250_NL01	2	66	-0.333	-0.203	14	1
connected_area_km2	250	B250_NO01	1	67	-0.329	-0.195	14	1
connected_area_km2	250	B250_UKBC01	5	63	-0.217	-0.125	13	3
connected_area_km2	250	B250_UKBC02	1	67	-0.315	-0.190	14	1
connected_area_km2	250	B250_UKC01	3	65	-0.331	-0.201	13	0
connected_area_km2	250	B250_UKC02	2	66	-0.346	-0.210	14	1
connected_area_km2	250	B250_UKI01	2	66	-0.362	-0.221	14	1
connected_area_km2	250	B250_UKI02	7	61	-0.310	-0.197	13	2
connected_area_km2	250	B250_UKI03	3	65	-0.344	-0.209	13	0
connected_area_km2	250	B250_UKNE01	2	66	-0.339	-0.205	14	1
connected_area_km2	250	B250_UKNE02	1	67	-0.332	-0.200	14	1
connected_area_km2	250	B250_UKNE03	3	65	-0.366	-0.220	13	0
connected_area_km2	250	B250_UKNE04	4	64	-0.326	-0.195	13	0
connected_area_km2	250	B250_UKSE01	3	65	-0.411	-0.263	13	0
connected_area_km2	250	B250_UKSW01	3	65	-0.352	-0.213	13	0
connected_area_km2	250	B250_UKWA01	4	64	-0.356	-0.225	13	0
connected_area_km2	250	B250_UKWA02	3	65	-0.345	-0.208	13	0

Supplementary Table S11. Coastal-basin-specific descriptive associations.

coastal sector	terrain metric	terrain metric label	n sites	spearman	kendall	status
Bristol-Channel	below_reference_land_pct	lowland_prevalence	6	0.543	0.467	estimated
Brittany-Atlantic	below_reference_land_pct	lowland_prevalence	4	-0.800	-0.667	estimated

coastal sector	terrain metric	terrain metric label	n sites	spearman	kendall	status
Celtic-Irish-Sea	below_reference_land_pct	lowland_prevalence	2	NA	NA	insufficient_n_lt4
Danish-Wadden-Coast	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
Dutch-German-Bight	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
Dutch-Wadden-Coast	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
English-Channel-UK	below_reference_land_pct	lowland_prevalence	5	-0.100	0.000	estimated
French-English-Channel	below_reference_land_pct	lowland_prevalence	9	-0.783	-0.556	estimated
German-Bight	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
Irish-Sea	below_reference_land_pct	lowland_prevalence	7	-0.786	-0.714	estimated
Loire-Bay-of-Biscay	below_reference_land_pct	lowland_prevalence	3	NA	NA	insufficient_n_lt4
North-Channel-Clyde	below_reference_land_pct	lowland_prevalence	3	NA	NA	insufficient_n_lt4
Northern-North-Sea	below_reference_land_pct	lowland_prevalence	7	-0.679	-0.524	estimated
Rhine-Meuse-Coast	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
Scottish-Atlantic	below_reference_land_pct	lowland_prevalence	7	-0.607	-0.333	estimated
Skagerrak	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
Southern-North-Sea-UK	below_reference_land_pct	lowland_prevalence	5	0.500	0.200	estimated
Southwest-Atlantic	below_reference_land_pct	lowland_prevalence	3	NA	NA	insufficient_n_lt4
Thames-Estuary	below_reference_land_pct	lowland_prevalence	1	NA	NA	insufficient_n_lt4
Bristol-Channel	conditional_connectivity_pct	conditional_connectivity	6	-0.319	-0.276	estimated
Brittany-Atlantic	conditional_connectivity_pct	conditional_connectivity	4	-1.000	-1.000	estimated
Celtic-Irish-Sea	conditional_connectivity_pct	conditional_connectivity	2	NA	NA	insufficient_n_lt4
Danish-Wadden-Coast	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
Dutch-German-Bight	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
Dutch-Wadden-Coast	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
English-Channel-UK	conditional_connectivity_pct	conditional_connectivity	5	0.500	0.400	estimated

coastal sector	terrain metric	terrain metric label	n sites	spearman	kendall	status
French-English-Channel	conditional_connectivity_pct	conditional_connectivity	9	-0.017	0.056	estimated
German-Bight	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
Irish-Sea	conditional_connectivity_pct	conditional_connectivity	7	0.072	0.098	estimated
Loire-Bay-of-Biscay	conditional_connectivity_pct	conditional_connectivity	3	NA	NA	insufficient_n_lt4
North-Channel-Clyde	conditional_connectivity_pct	conditional_connectivity	3	NA	NA	insufficient_n_lt4
Northern-North-Sea	conditional_connectivity_pct	conditional_connectivity	7	-0.821	-0.619	estimated
Rhine-Meuse-Coast	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
Scottish-Atlantic	conditional_connectivity_pct	conditional_connectivity	7	0.071	0.143	estimated
Skagerrak	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
Southern-North-Sea-UK	conditional_connectivity_pct	conditional_connectivity	5	-0.400	-0.400	estimated
Southwest-Atlantic	conditional_connectivity_pct	conditional_connectivity	3	NA	NA	insufficient_n_lt4
Thames-Estuary	conditional_connectivity_pct	conditional_connectivity	1	NA	NA	insufficient_n_lt4
Bristol-Channel	connected_reference_land_pct	connected_lowland_share	6	0.543	0.467	estimated
Brittany-Atlantic	connected_reference_land_pct	connected_lowland_share	4	-1.000	-1.000	estimated
Celtic-Irish-Sea	connected_reference_land_pct	connected_lowland_share	2	NA	NA	insufficient_n_lt4
Danish-Wadden-Coast	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
Dutch-German-Bight	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
Dutch-Wadden-Coast	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
English-Channel-UK	connected_reference_land_pct	connected_lowland_share	5	-0.100	0.000	estimated
French-English-Channel	connected_reference_land_pct	connected_lowland_share	9	-0.817	-0.611	estimated
German-Bight	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
Irish-Sea	connected_reference_land_pct	connected_lowland_share	7	-0.714	-0.524	estimated
Loire-Bay-of-Biscay	connected_reference_land_pct	connected_lowland_share	3	NA	NA	insufficient_n_lt4
North-Channel-Clyde	connected_reference_land_pct	connected_lowland_share	3	NA	NA	insufficient_n_lt4

coastal sector	terrain metric	terrain metric label	n sites	spearman	kendall	status
Northern-North-Sea	connected_reference_land_pct	connected_lowland_share	7	-0.679	-0.524	estimated
Rhine-Meuse-Coast	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
Scottish-Atlantic	connected_reference_land_pct	connected_lowland_share	7	-0.643	-0.429	estimated
Skagerrak	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
Southern-North-Sea-UK	connected_reference_land_pct	connected_lowland_share	5	0.100	0.000	estimated
Southwest-Atlantic	connected_reference_land_pct	connected_lowland_share	3	NA	NA	insufficient_n_lt4
Thames-Estuary	connected_reference_land_pct	connected_lowland_share	1	NA	NA	insufficient_n_lt4
Bristol-Channel	connected_area_km2	connected_area	6	0.771	0.600	estimated
Brittany-Atlantic	connected_area_km2	connected_area	4	-0.600	-0.333	estimated
Celtic-Irish-Sea	connected_area_km2	connected_area	2	NA	NA	insufficient_n_lt4
Danish-Wadden-Coast	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
Dutch-German-Bight	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
Dutch-Wadden-Coast	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
English-Channel-UK	connected_area_km2	connected_area	5	-0.100	0.000	estimated
French-English-Channel	connected_area_km2	connected_area	9	-0.817	-0.611	estimated
German-Bight	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
Irish-Sea	connected_area_km2	connected_area	7	-0.643	-0.429	estimated
Loire-Bay-of-Biscay	connected_area_km2	connected_area	3	NA	NA	insufficient_n_lt4
North-Channel-Clyde	connected_area_km2	connected_area	3	NA	NA	insufficient_n_lt4
Northern-North-Sea	connected_area_km2	connected_area	7	-0.750	-0.619	estimated
Rhine-Meuse-Coast	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
Scottish-Atlantic	connected_area_km2	connected_area	7	-0.464	-0.333	estimated
Skagerrak	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
Southern-North-Sea-UK	connected_area_km2	connected_area	5	0.200	0.200	estimated

coastal sector	terrain metric	terrain metric label	n sites	spearman	kendall	status
Southwest-Atlantic	connected_area_km2	connected_area	3	NA	NA	insufficient_n_lt4
Thames-Estuary	connected_area_km2	connected_area	1	NA	NA	insufficient_n_lt4
Bristol-Channel	connected_window_pct	connected_window_share	6	0.771	0.600	estimated
Brittany-Atlantic	connected_window_pct	connected_window_share	4	-0.600	-0.333	estimated
Celtic-Irish-Sea	connected_window_pct	connected_window_share	2	NA	NA	insufficient_n_lt4
Danish-Wadden-Coast	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4
Dutch-German-Bight	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4
Dutch-Wadden-Coast	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4
English-Channel-UK	connected_window_pct	connected_window_share	5	-0.100	0.000	estimated
French-English-Channel	connected_window_pct	connected_window_share	9	-0.817	-0.611	estimated
German-Bight	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4
Irish-Sea	connected_window_pct	connected_window_share	7	-0.643	-0.429	estimated
Loire-Bay-of-Biscay	connected_window_pct	connected_window_share	3	NA	NA	insufficient_n_lt4
North-Channel-Clyde	connected_window_pct	connected_window_share	3	NA	NA	insufficient_n_lt4
Northern-North-Sea	connected_window_pct	connected_window_share	7	-0.750	-0.619	estimated
Rhine-Meuse-Coast	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4
Scottish-Atlantic	connected_window_pct	connected_window_share	7	-0.464	-0.333	estimated
Skagerrak	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4
Southern-North-Sea-UK	connected_window_pct	connected_window_share	5	0.200	0.200	estimated
Southwest-Atlantic	connected_window_pct	connected_window_share	3	NA	NA	insufficient_n_lt4
Thames-Estuary	connected_window_pct	connected_window_share	1	NA	NA	insufficient_n_lt4

Supplementary Table S12. Moran spatial-autocorrelation diagnostics.

variable	distance threshold km	moran i	expected i under randomization	permutation two sided p	null q025	null q975	weight definition	n permutations
coast_rp_rp10_m	150.000	0.309	-0.015	0.001	-0.174	0.165	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
below_reference_land_pct	150.000	0.658	-0.015	4.000e-04	-0.132	0.176	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
conditional_connectivity_pct	150.000	-0.022	-0.015	0.979	-0.149	0.179	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
connected_reference_land_pct	150.000	0.438	-0.015	0.003	-0.116	0.176	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
connected_area_km2	150.000	0.495	-0.015	0.002	-0.102	0.159	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999

variable	distance threshold km	moran i	expected i under randomization	permutation two sided p	null q025	null q975	weight definition	n permutations
connected_window_pct	150.000	0.494	-0.015	0.002	-0.102	0.162	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
coast_rp_rp10_m	250.000	0.115	-0.015	0.038	-0.112	0.106	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
below_reference_land_pct	250.000	0.543	-0.015	4.000e-04	-0.093	0.112	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
conditional_connectivity_pct	250.000	0.018	-0.015	0.435	-0.103	0.108	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
connected_reference_land_pct	250.000	0.417	-0.015	0.001	-0.084	0.099	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999
connected_area_km2	250.000	0.389	-0.015	4.000e-04	-0.076	0.083	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999

variable	distance threshold km	moran i	expected i under randomization	permutation two sided p	null q025	null q975	weight definition	n permutations
connected_window_pct	250.000	0.389	-0.015	8.000e-04	-0.077	0.084	row-standardized symmetric binary great-circle distance; isolates linked symmetrically to nearest neighbour	4999

Supplementary Table S13. Maximum leave-one-station influence by terrain metric.

Terrain metric	Most influential omitted station	Coastal grouping	Full rho	Leave-one-out rho	Change in rho
lowland_prevalence	Hinkleypoint P003 Uk	Bristol-Channel	-0.309	-0.279	0.030
conditional_connectivity	Hinkleypoint P003 Uk	Bristol-Channel	0.105	0.070	-0.035
connected_area	Sheerness	Thames-Estuary	-0.341	-0.372	-0.031
connected_lowland_share	Sheerness	Thames-Estuary	-0.333	-0.364	-0.031
connected_window_share	Sheerness	Thames-Estuary	-0.341	-0.372	-0.031

Supplementary Table S14. Explicit water-seed and land-graph topology summary.

water neighbours	land neighbours	terrain metric	n sites	water vs terrain spearman	terrain rank spearman vs primary 8water 4land	top20 n	top20 overlap with primary	top20 membership changes
4	4	connected_reference_land_pct	68	-0.427	0.862	14	12	4
4	4	connected_area_km2	68	-0.431	0.862	14	12	4
4	4	conditional_connectivity_pct	68	0.089	0.957	14	14	0
4	8	connected_reference_land_pct	68	-0.366	0.888	14	11	6
4	8	connected_area_km2	68	-0.371	0.896	14	11	6
4	8	conditional_connectivity_pct	68	0.012	0.867	14	13	2
8	4	connected_reference_land_pct	68	-0.333	1.000	14	14	0
8	4	connected_area_km2	68	-0.341	1.000	14	14	0
8	4	conditional_connectivity_pct	68	0.105	1.000	14	14	0

water neighbours	land neighbours	terrain metric	n sites	water vs terrain spearman	terrain rank spearman vs primary 8water 4land	top20 n	top20 overlap with primary	top20 membership changes
8	8	connected_reference_land_pct	68	-0.320	0.961	14	11	6
8	8	connected_area_km2	68	-0.329	0.967	14	11	6
8	8	conditional_connectivity_pct	68	-0.006	0.901	14	13	2

Supplementary Table S15. Decision-level rank displacement and terrain-selection frequency.

Station	Water rank	Terrain rank	Absolute rank displacement	Primary priority class	Eligible configurations	Selected configurations	Terrain top-20% selection frequency
Hinkleypoint P003 Uk	4	68	64	water-priority-only	40	0	0.000
Newport P057 Uk	3	67	64	water-priority-only	40	0	0.000
Denhelder Hel Nl	58	2	56	terrain-priority-only	40	40	1.000
Ilfracombe P061 Uk	10	65	55	water-priority-only	40	0	0.000
Tregde 006 Norway	68	13	55	terrain-priority-only	40	30	0.750
Mumbles P932 Uk	9	63	54	water-priority-only	40	0	0.000
Sthelier P074 Uk	7	61	54	water-priority-only	40	0	0.000
Dieppe France	8	59	51	water-priority-only	40	0	0.000
Gladstone P234 Uk	12	62	50	water-priority-only	40	0	0.000
Portbury P341 Uk	2	50	48	water-priority-only	22	0	0.000
Avonmouth P060 Uk	1	48	47	water-priority-only	22	0	0.000
Saint Malo France	5	52	47	water-priority-only	40	0	0.000
Portsmouth P008 Uk	56	10	46	terrain-priority-only	40	40	1.000
Weymouth P991 Uk	64	19	45	neither-top20	40	2	0.050
Lerwick P041 Uk	65	21	44	neither-top20	40	0	0.000
Dover P012 Uk	20	58	38	neither-top20	40	0	0.000
Llandudno P007 Uk	15	51	36	neither-top20	40	0	0.000
Morayfirth P207 Uk	52	16	36	neither-top20	40	10	0.250

Station	Water rank	Terrain rank	Absolute rank displacement	Primary priority class	Eligible configurations	Selected configurations	Terrain top-20% selection frequency
Bangor P662 Uk	59	25	34	neither-top20	40	8	0.200
Lowestoft P024 Uk	62	28	34	neither-top20	40	5	0.125
Wick P035 Uk	60	26	34	neither-top20	40	4	0.100
Portellen P202 Uk	66	34	32	neither-top20	40	0	0.000
Brest	24	55	31	neither-top20	40	0	0.000
Malin Head 834a	61	31	30	neither-top20	40	5	0.125
Millport P049 Uk	57	27	30	neither-top20	40	0	0.000
Stornoway P042 Uk	53	23	30	neither-top20	40	0	0.000
Saint Gildas France	44	15	29	neither-top20	40	17	0.425
Stmarys P231 Uk	38	9	29	terrain-priority-only	40	40	1.000
Immingham P026 Uk	22	49	27	neither-top20	22	10	0.455
Le Conquet France	27	54	27	neither-top20	40	0	0.000
Leith P034 Uk	33	60	27	neither-top20	40	0	0.000
Workington P217 Uk	13	40	27	water-priority-only	40	0	0.000
Bournemouth P988 Uk	67	41	26	neither-top20	40	2	0.050
Heysham P050 Uk	6	32	26	water-priority-only	40	0	0.000
Holyhead P054 Uk	40	14	26	terrain-priority-only	40	14	0.350
Boulogne Sur Mer	11	36	25	water-priority-only	40	3	0.075
Delfzijl Del Nl	26	1	25	terrain-priority-only	40	40	1.000
Esbjerg 130121 Denmark	36	11	25	terrain-priority-only	40	35	0.875
Northshields P032 Uk	41	66	25	neither-top20	40	0	0.000
Portrush P935 Uk	63	38	25	neither-top20	40	0	0.000
Cromer P183 Uk	30	53	23	neither-top20	40	0	0.000
Milfordhaven P056 Uk	23	46	23	neither-top20	40	0	0.000
Roscoff France	16	39	23	neither-top20	40	0	0.000
Aberdeen	55	33	22	neither-top20	40	0	0.000
Barmouth P923 Uk	34	12	22	terrain-priority-only	40	23	0.575

Station	Water rank	Terrain rank	Absolute rank displacement	Primary priority class	Eligible configurations	Selected configurations	Terrain top-20% selection frequency
Felixstowe P204 Uk	29	7	22	terrain-priority-only	40	34	0.850
Le Crouesty France	46	24	22	neither-top20	40	0	0.000
Whitby P174 Uk	35	57	22	neither-top20	40	0	0.000
Harwich P022 Uk	28	8	20	terrain-priority-only	40	40	1.000
Port Tudy France	50	30	20	neither-top20	40	9	0.225
Hoek van Holland	37	20	17	neither-top20	40	21	0.525
Dunkerque France	21	5	16	terrain-priority-only	40	12	0.300
Cuxhaven Germany	18	3	15	terrain-priority-only	40	22	0.550
Concarneau France	49	35	14	neither-top20	40	0	0.000
Fishguard P055 Uk	51	64	13	neither-top20	40	0	0.000
Ullapool P043 Uk	43	56	13	neither-top20	40	0	0.000
Calais France	17	6	11	terrain-priority-only	40	20	0.500
Devonport P002 Uk	48	37	11	neither-top20	40	0	0.000
Porterin P919 Uk	32	22	10	neither-top20	40	2	0.050
Portpatrick P063 Uk	54	44	10	neither-top20	40	0	0.000
Sheerness	14	4	10	dual-priority	40	40	1.000
Newhaven P011 Uk	25	17	8	neither-top20	40	0	0.000
Newlyn	39	47	8	neither-top20	40	0	0.000
Kinlochbervi P918 Uk	45	42	3	neither-top20	40	0	0.000
Cherbourg France	31	29	2	neither-top20	40	0	0.000
Tobermory P223 Uk	47	45	2	neither-top20	40	0	0.000
Le Havre France	19	18	1	neither-top20	40	14	0.350
Saint Nazaire France	42	43	1	neither-top20	40	0	0.000

Supplementary Table S16. Recorded Figure 3 case-selection rules and values.

Station	Figure 3 role	Recorded selection rule	RP10 (m)	Water rank	Terrain rank	Below-2 m share (%)	Conditional connection (%)	Connected share (%)	Connected area (km2)
Delfzijl Del Nl	stable terrain-priority case	Top-20% connected-land share, local-DTM IoU >= 0.75; smallest absolute local-minus-DeltaDTM share difference	4.094	26	1	97.003	99.692	96.704	61.303
Newport P057 Uk	water-priority, lowland-limited case	Top-20% storm tide and conditional connectivity >= 90%; maximum terrain-minus-water rank displacement, then highest water rank	8.602	3	67	0.164	90.528	0.148	0.102
Immingham P026 Uk	connectivity-filtered case	Below-2 m land share at or above the sample median; minimum conditional connectivity	4.528	22	49	20.142	2.014	0.406	0.249

Supplementary Table S17. Seven-site local-DTM product, datum and resampling cross-check.

Station	Local terrain product	Native datum	Delta connected share (%)	Local connected share (%)	Delta conditional connection (%)	Local conditional connection (%)	Delta area (km2)	Local area (km2)	Connected-mask IoU	Elevation resampling
Sheerness	Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2	Ordnance Datum Newlyn / EA product native datum	42.700	9.630	93.412	33.626	20.364	4.612	0.215	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service

Station	Local terrain product	Native datum	Delta connected share (%)	Local connected share (%)	Delta conditional connection (%)	Local conditional connection (%)	Delta area (km2)	Local area (km2)	Connected-mask IoU	Elevation resampling
Newlyn	Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2	Ordnance Datum Newlyn / EA product native datum	0.446	0.417	98.367	98.931	0.276	0.259	0.528	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service
Lowestoft P024 Uk	Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2	Ordnance Datum Newlyn / EA product native datum	1.416	6.165	11.707	46.234	0.670	2.928	0.111	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service
Immingham P026 Uk	Environment Agency LiDAR Composite DTM 2 m WCS, scaleFactor 0.2	Ordnance Datum Newlyn / EA product native datum	0.406	0.079	2.014	0.819	0.249	0.049	0.042	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service
Denhelder Hel Nl	PDOK/Rijkswaterstaat AHN DTM 0.5 m WCS, scaleFactor 0.05	Normaal Amsterdams Peil / AHN product native datum	71.730	4.365	98.327	6.770	20.855	1.274	0.055	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service
Delfzijl Del Nl	PDOK/Rijkswaterstaat AHN DTM 0.5 m WCS, scaleFactor 0.05	Normaal Amsterdams Peil / AHN product native datum	96.704	89.784	99.692	99.513	61.303	57.187	0.926	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service

Station	Local terrain product	Native datum	Delta connected share (%)	Local connected share (%)	Delta conditional connection (%)	Local conditional connection (%)	Delta area (km2)	Local area (km2)	Connected-mask IoU	Elevation resampling
Hoek van Holland	PDOK/Rijkswaterstaat AHN DTM 0.5 m WCS, scaleFactor 0.05	Normaal Amsterdams Peil / AHN product native datum	2.640	1.439	5.484	3.537	1.411	0.772	0.217	provider WCS scaleFactor to nominal 10 m; interpolation kernel controlled by provider service

Supplementary Table S18. Excluded COAST-RP nearest-point matches.

Station	Latitude	Longitude	COAST-RP point	Distance (km)	Reason
Cordemais France	47.277	-1.890	id_obs_donges_donges_france_refmar	22.647	excluded_nearest_coast_rp_point_gt6km
Le Pellerin France	47.205	-1.765	IBIS_GL_TS_MO_IF000578	28.388	excluded_nearest_coast_rp_point_gt6km
Nantes Usine Brulee	47.204	-1.574	IBIS_GL_TS_MO_IF000578	39.423	excluded_nearest_coast_rp_point_gt6km
Donges France	47.306	-2.088	id_obs_donges_donges_france_refmar	8.278	excluded_nearest_coast_rp_point_gt6km
Montoir De Bretagne	47.306	-2.108	id_obs_donges_donges_france_refmar	6.897	excluded_nearest_coast_rp_point_gt6km
Paimboeuf France	47.287	-2.000	id_obs_donges_donges_france_refmar	14.337	excluded_nearest_coast_rp_point_gt6km

Supplementary Table S19. Focal GSSR and COAST-RP product-context diagnostics.

Station	GSSR empirical RP10 (m)	Bootstrap 2.5%	Bootstrap 97.5%	GSSR lower-bound RP10	GSSR upper-bound RP10	COAST-RP RP10 (m)	Within-5 km minimum	Within-5 km maximum	D10 cross-product difference (m)	Tidal range (m)	Tidal source status
Sheerness	1.363	1.277	1.470	0.999	1.705	5.256	5.085	5.256	3.893	5.200	source_verified
Newlyn	0.617	0.563	0.673	0.469	0.766	3.346	3.346	3.348	2.729	4.600	source_verified
Aberdeen	0.838	0.789	0.882	0.663	1.014	2.822	2.816	2.822	1.984	3.500	source_verified
Hoek van Holland	1.505	1.430	1.622	1.204	1.810	3.373	3.360	3.373	1.868	1.700	indicative_context
Brest	0.687	0.645	0.847	0.517	0.857	4.313	4.233	4.321	3.626	6.100	indicative_context
New York	1.295	1.067	1.459	1.064	1.525	1.311	1.311	1.311	0.016	1.380	source_verified
Hong Kong	1.026	0.751	1.143	0.829	1.225	1.823	1.823	1.823	0.797	1.900	official_general_context
Charleston	0.843	0.731	1.259	0.630	1.069	1.346	1.346	1.356	0.503	1.591	source_verified

Supplementary Table S20. Source-verified tidal stations plotted in Supplementary Figure S1.

Station	Spring tidal range (m)	COAST-RP RP10 (m)	Connected land share (%)	Source	URL
Newlyn	4.600	3.346	0.446	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Aberdeen	3.500	2.822	0.982	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Sheerness	5.200	5.256	42.700	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Lerwick P041 Uk	1.600	1.508	2.338	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Portpatrick P063 Uk	3.400	2.887	0.546	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Fishguard P055 Uk	3.700	2.998	0.178	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Milfordhaven P056 Uk	6.100	4.376	0.472	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Lowestoft P024 Uk	1.800	2.379	1.416	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Wick P035 Uk	2.700	2.560	1.552	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Stornoway P042 Uk	3.900	2.980	1.911	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Immingham P026 Uk	6.000	4.528	0.406	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo
Millport P049 Uk	2.800	2.717	1.537	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo

Station	Spring tidal range (m)	COAST-RP RP10 (m)	Connected land share (%)	Source	URL
Holyhead P054 Uk	4.800	3.315	5.703	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo

Supplementary Table S21. Complete tidal metadata source audit.

Station	Metric	Range (m)	Source status	Source	URL	Notes
Newlyn	mean spring range = MHWS - MLWS	4.600	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Hoek van Holland	spring range context	1.700	indicative_context	Port of Rotterdam/Rijkswaterstaat candidate source identified; exact station spring range not yet extracted automatically	https://www.portofrotterdam.com/nl/actuele-informatie/weer-getijde-en-waterdiepte	Retain as context until HydroMeteoBundel or Rijkswaterstaat station statistics are parsed.
Brest	spring range context	6.100	indicative_context	SHOM/REFMAR tidal-prediction definitions identified; exact station spring range not yet extracted automatically	https://refmar.shom.fr/sites/default/files/2025-01/GT-TSH_CatD_Fiche_Prediction_maree.pdf	Retain as context until SHOM station PMVE/BMVE or harmonic metadata are parsed.
Aberdeen	mean spring range = MHWS - MLWS	3.500	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Sheerness	mean spring range = MHWS - MLWS	5.200	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Esbjerg 130121 Denmark	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Denhelder Hel Nl	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Lerwick P041 Uk	mean spring range = MHWS - MLWS	1.600	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.

Station	Metric	Range (m)	Source status	Source	URL	Notes
Cuxhaven Germany	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Delfzijl Del Nl	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Portpatrick P063 Uk	mean spring range = MHWS - MLWS	3.400	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Fishguard P055 Uk	mean spring range = MHWS - MLWS	3.700	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Milfordhaven P056 Uk	mean spring range = MHWS - MLWS	6.100	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Tregde 006 Norway	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Lowestoft P024 Uk	mean spring range = MHWS - MLWS	1.800	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Wick P035 Uk	mean spring range = MHWS - MLWS	2.700	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Stornoway P042 Uk	mean spring range = MHWS - MLWS	3.900	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Le Conquet France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.

Station	Metric	Range (m)	Source status	Source	URL	Notes
Port Tudy France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Cherbourg France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Saint Nazaire France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Saint Gildas France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Le Havre France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Dunkerque France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Roscoff France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Boulogne Sur Mer	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.
Dieppe France	not available	NA	missing_or_not_yet_extracted	NA	NA	Non-UK/non-US extended-site tidal metadata not automatically extracted in this run.

Station	Metric	Range (m)	Source status	Source	URL	Notes
Immingham P026 Uk	mean spring range = MHWS - MLWS	6.000	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Millport P049 Uk	mean spring range = MHWS - MLWS	2.800	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.
Holyhead P054 Uk	mean spring range = MHWS - MLWS	4.800	source_verified	National Tidal and Sea Level Facility highest/lowest predicted tides table	https://ntslf.org/tides/uk-network/predictions-hilo	Values parsed from the NTSLF AJAX table; heights are given relative to chart datum.

Supplementary Table S22. GSSR metadata-field and filter audit.

audit item	value	interpretation
available_metadata_fields	FIELD1;corrn;num_95pcs;num_year;path;rmse;tg	corrn is retained as the source metadata reconstruction correlation; no separate extreme-event skill field is present in the local metadata.
eligible_corr_ge0.45_years_ge20	62	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.45_years_ge25	47	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.45_years_ge30	38	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.55_years_ge20	61	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.55_years_ge25	46	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.55_years_ge30	38	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.65_years_ge20	59	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.65_years_ge25	45	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.
eligible_corr_ge0.65_years_ge30	38	Supporting GSSR-quality subset count only; this filter does not define the primary COAST-RP-terrain sample.

Supplementary Table S23. Threshold-integrated terrain associations (0-3 m EGM2008).

terrain metric	n sites	spearman	kendall	interpretation
connected_share_auc_0_3m_pct	68	-0.298	-0.187	threshold-integrated standardized EGM2008 terrain screen; not a flood probability
connected_area_auc_0_3m_km2	68	-0.313	-0.186	threshold-integrated standardized EGM2008 terrain screen; not a flood probability
below_share_auc_0_3m_pct	68	-0.289	-0.176	threshold-integrated standardized EGM2008 terrain screen; not a flood probability

Supplementary Table S24. Coastline-oriented versus square-window sensitivity.

terrain metric	n sites	terrain rank spearman oriented vs square	water vs oriented spearman	water vs square spearman	top20 n	top20 overlap oriented vs square	top20 membership changes	interpretation
connected_reference_land_pct	68	0.804	-0.248	-0.333	14	11	6	mask-derived window-geometry sensitivity, not an alternative flood model
connected_area_km2	68	0.742	-0.254	-0.341	14	11	6	mask-derived window-geometry sensitivity, not an alternative flood model
below_reference_land_pct	68	0.893	-0.202	-0.309	14	14	0	mask-derived window-geometry sensitivity, not an alternative flood model

Supplementary Table S25. COAST-RP coordinate-connector review.

station_id	station	tg	lat	lon	n_my_ear	coorn	rmse	gssr_archive	archive_available	pass_esy_ear_sge_25	pass_esscoorr_geo_45	pass_esscoorr_geo_55	pass_esscoorr_geo_65	selected_base_30	selection_reason	match_dist_km	coast_rp_10m	delta_add_tm_tile_available	delta_add_tm_tiles	coast_rp_ssour_ce_station_id	coast_rp_ssour_ce_lat	coast_rp_ssour_ce_lon	recomputed_match_dist_km	included_in_primary_match_sample	match_exclusion_reason	threshold_note	coast_albasin	spatial_block_100km	spatial_block_150km	spatial_block_250km	block_definition	coastal_sector	coastal_region	source_id_contains_station_name	estuary_or_embayment_flag	island_fjord_geometry_flag	opposite_bank_flag	coord_in_at_econnector_review_pass	review_status	review_basis

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| cuxhaven-germany | CuxhavenGermany | cuxhaven-germany-bsh.csv | 53.867 | 8.717 | 37 | 0.821 | 0.234 | cuxhaven_germany.7z | True | True | True | True | True | True | True | included_in_baseline_30-stations_subset | 0.531 | 4.696 | True | DeltaDTM_v1_1_N53E008.tif | id_coast_glob_eur_03989 | 53.870 | 8.723 | 0.531 | True | included_match_le6km | The6km rule is a pre-specified extraction screen, not an along-coast representa | German-Bight | B100_DE01 | B150_DE01 | B250_DE01 | reviewed-coastline-v2 | German-Bight | South-Northern-Sea-Continental | False | True | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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wick-po35-uk	WickPo35Uk	wick-po35-uk-bodc.csv	58.441	-3.086	36	0.835	0.089	wick_po35_uk.7z	True	True	True	True	True	True	True	included in baseline 30-station subset	2.044	2.560	True	DeltaDTM_v1_1_N58Woo4.tif	IBIS_IR_TS_MO_Wick	58.425	-3.069	2.044	True	included_match_le6km	The 6km rule is a pre-specified extraction screen, not an along-coast representa
Northern-North-Sea	B100_UKNEoo6	B150_UKNEoo5	B250_UKNEoo4	reviewed-coastline-v2	Northern-North-Sea	British-North-Sea	True	False	False	not_determined_from_coordinate_review	True	coordinate_plausible_not_hydrodynamic_equivalence	coordinate connection inspection; source-station identifier whereas named; ex

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| le-conquet | LeConquetFrance | le_conquet-france-ref mar.csv | 48.359 | -4.781 | 36 | 0.808 | 0.076 | le_conquet_.7z | True | True | True | True | True | True | included in base line 30-station subset | 0.374 | 4.027 | True | DeltaDTM_v1_1_N48W005.tif | IBIS_GL_TS_MO_LeConquetTG | 48.362 | -4.783 | 0.374 | True | included_match_le6km | The 6km ruleis a pre-specified extraction screen, not a along-coast representa | Brittany-Atlantic | B100_FRB03 | B150_FRB02 | B250_FRB02 | reviewed-coastline-v2 | Brittany-Atlantic | French-Atlantic-and-Brittany | False | False | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier whereas named; ex |
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| brest | Brest | brest-france-refmar.csv | 48.383 | -4.500 | 36 | 0.795 | 0.084 | brest_.7z | True | True | True | True | True | True | included_in_base_line_30-stations_subset | 0.833 | 43.13 | True | DeltaDTM_v1_1_N48W005.tif | id_obs_brest_822a_france_uhslc | 48.3776 | -4.504 | 0.833 | True | included_match_le6km | The6km_rule_is_a_pre-specified_extraction_screen,_not_a_along-coast_representa | Brittany-Atlantic | B1oo_FRBo3 | B150_FRBo2 | B250_FRBo2 | reviewed-coastline-v2 | Brittany-Atlantic | French-Atlantic-and-Brittany | True | False | False | not_determined_from_coordinate_review | True | coordinate_pair_plausible_not_hydrodynamic_equivalence | coordinate_connector_inspection;_source-station_identifier_where_renamed;_ex |

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| port-study | PortStudyFrance | port-study-france-ref | 47.6444 | -34.446 | 36 | 0.790 | 0.078 | port_study_.7z | True | True | True | True | True | True | True | included_in_base_line_30-station_subset | 0.307 | 3.000 | True | DeltaDTM_v1_1_N47Woo4.tif | IBIS_GL_TS_MO_PortTudyTG | 47.6444 | -3.450 | 0.307 | True | included_match_le6km | The6kmruleisapre-specifiedextractionscreen,notanalongg-coastrepresenta | Brittany-Atlantic | B100_FRBo2 | B150_FRBo1 | B250_FRBo1 | reviewed-coastline-v2 | Brittany-Atlantic | French-Atlantic-and-Brittany | False | False | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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| cherbourg | Cherbourg France | cherbourg-france-refmar.csv | 49.651 | -1.635 | 36 | 0.785 | 0.089 | cherbourg_.7z | True | True | True | True | True | True | included in base line 30-station subset | 0.180 | 36.74 | True | DeltaDTM_v1_1_N49W002.tif | id_obs_cherbourg_cherbourg_france_refmar | 49.651 | -1.633 | 0.180 | True | included_match_le6km | The 6km rule is applied exactly as specified: extraction screens, not analog-coastrepresenta | French_Engl_FRCO2 | B150_FRCO2 | B250_FRCO1 | reviewed-coastline-v2 | French-English-Channel | French-English-Channel | True | False | False | not_determined_from_coordinate_review | True | coordinate_pair_plausible_not_hydrodynamic_equivalence | coordinate connection; source-station identifier where renamed; ex |
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| saint-nazaire | SaintNazaireFrance | saint-nazaire-france-refmar.csv | 47.267 | -2.202 | 36 | 0.776 | 0.106 | saint_nazaire_.7z | True | True | True | True | True | True | included_in_base_line_30-station_subset | 0.513 | 3.275 | True | DeltaDTM_v1_1_N47W003.tif | id_obs_saint_nazaire_franc | 47.263 | -2.205 | 0.513 | True | included_match_le6km | The6kmrulisapre-specifiedextractionscreen,notanalongng-coastrepresenta | Loire-Bay-of-Biscay | B1oo_FRL01 | B15o_FRL01 | B25o_FRL01 | reviewed-coastline-v2 | Loire-Bay-of-Biscay | French-Atlantic-and-Brittany | False | False | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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| saint-gildas | SaintGildasFrance | saint-gildas-france-refmar.csv | 47.140 | -2.246 | 36 | 0.773 | 0.095 | saint_gildas_.7z | True | True | True | True | True | True | True | included_in_base_line_30-stations_subset | 0.696 | 3.243 | True | DeltaDTM_v1_1_N47W003.tif | id_obs_saint_gildas_saint_gildas_france_ | 47.1446 | -2.249 | 0.696 | True | included_match_le6km | The6kmrulesisapre-specifiedextractionscreen,notanalongong-coastrepresenta | Loire-Bay-of-Biscay | B100_FRL01 | B150_FRL01 | B250_FRL01 | reviewed-coastline-v2 | Loire-Bay-of-Biscay | French-Atlantic-and-Brittany | True | False | False | not_determined_from_coordinate_review | True | coordinate_pair_plausible_not_hydrodynamic_equivalence | coordinate connection; source-station identifier where renamed; ex |
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| le-have | LeHaveFrance | le_have-france-reform ar.csv | 49.482 | 0.106 | 36 | 0.755 | 0.113 | le_have_.7z | True | True | True | True | True | True | True | included in base line 30-station subset | 1.113 | 4.643 | True | DeltaDTM_v1_1_N49E000.tif | id_coast_glob_eur_03862 | 49.475 | 0.095 | 1.113 | True | included_match_le6km | The 6km rule is a pre-specified extraction screen, not an along-coast representa | French_FRCO3 | B1500_FRCO3 | B2500_FRCO2 | reviewed-coastline-v2 | French-English-Channel | French-English-Channel | False | False | False | not_determined_from_coordinate_review | True | coordinate_partial_possible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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dunkerque	Dunkerque France	dunkerque-france-refmar.csv	51.048	2.367	36	0.734	0.153	dunkerque_.7z	True	True	True	True	True	True	included in base line 30-station subset	1.002	4.548	True	Delta DTM_v1_1_N51E002.tif	id_coast_glob_eur_03898	51.057	2.366	1.002	True	included_match_le6km	The 6km rule is a pre-specified extraction screen, not an along-coast representa
French-English-Channel	B1005_FRCO5	B1505_FRCO4	B2505_FRCO3	reviewed-coastline-v2	French-English-Channel	French-English-Channel	False	False	False	not_determined_from_coordinate_review	True	coordinate_plausible_not_hydrodynamic_equivalence	coordinate connection; source-station identifier whereas named; ex

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| dongess | DongessFrance | dongess-france-refmar.csv | 47.3006 | -20.088 | 26 | 0.789 | 0.112 | dongess_.7z | True | True | True | True | True | False | exclude d:passedfilterbutoutsidetop-30fillafterretaininginfocalcoresites | 8.2778 | 3.276 | True | DeltaDTM_v1_1_N47W003.tif | id_obs_dongess_france_refmar | 47.2778 | -2.190 | 8.278 | False | excluded_nearest_coast_rp_point_gt6km | The6kmruleisapre-specifiedextractionscreen,notanalongng-coastrepresenta | Loire-Estuary | B100_FRL02 | B150_FRL01 | B250_FRL01 | reviewed-coastline-v2 | Loire-Estuary | French-Atlantic-and-Brittany | True | False | False | not_determined_from_coordinate_review | False | excluded_distance_gt6km | coordinateconnectorinspection;sourceidentifierwhereenamed;ex |

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| portellen-p2022-uk | PortellenP2022Uk | portellen-p2022-uk-bodc.csv | 55.6270 | -6.190 | 20 | 0.820 | 0.114 | portellen_p2022_uk.7z | True | False | True | True | True | False | excluded:fewerthan25metadatayears | 0.069 | 1.437 | True | DeltaDTM_v1_1_N55W007.tif | IBIS_IR_TS_MO_PortEllen | 55.6270 | -6.189 | 0.069 | True | included_match_le6km | The6kmruleisapre-specifiedextractionscreen,notanalongs-coastrepresentativeUKWAO1 | Scottish-Atlantic | B1000_UKWAO1 | B1500_UKWAO1 | B2500_UKWAO1 | reviewed-coastline-v2 | Scottish-Atlantic | Irish-Sea-and-Scottish-Atlantic | True | False | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |

tiveness guarantee.

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| saint-malo | Saint Malo France | saint-malo-france-reform ar.csv | 48.6411 | -2.0278 | 20 | 0.725 | 0.100 | saint_malo_.7z | True | False | True | True | True | False | excluded: fewer than 25 metadata years | 0.129 | 7.4288 | True | DeltaDTM_v1_1_N48W003.tif; DeltaDTM_v1_1_N48W002.tif | id_obs_saint_malo_saint_malo_france_reform | 48.6400 | -2.029 | 0.129 | True | included_match_le6km | The 6km rule is a pre-specified extraction screen, not an analog-coast representa | French Polynesia | BR500_FRCO1 | BR500_FRCO1 | BR250_FRCO1 | reviewed-coastline-v2 | French-English-Channel | French-English-Channel | True | False | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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| bournemouth-p988-uk | Bournemouth P988 Uk | bournemouth-p988-uk-bodc.csv | 50.714 | -1.875 | 17 | 0.750 | 0.086 | bournemouth_p988_uk.7z | True | False | True | True | True | False | exclude d: fewer than 25 metadata years | 0.825 | 1.395 | True | Delta DTM_v1_1_N50 Woo2.tif | IBIS_IR_TS_MO_Bournemouth | 50.720 | -1.868 | 0.825 | True | included_match_le 6km | The 6km rule is a pre-specified extraction screen, not a along-coast representa | English-Channel-UK | B100_UKCO1 | B150_UKCO1 | B250_UKCO1 | reviewed-coastline-v2 | English-Channel-UK | English-Channel-UK | True | False | False | not_determined_from_coordinate_review | True | coordinate_pair_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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| le-crouesty | LeCrouestyFrance | le_crouesty-france-refmar.csv | 47.5443 | -2.8955 | 12 | 0.793 | 0.087 | le_crouesty_.7z | True | False | True | True | True | False | excluded:fewerthan25metadatayears | 0.199 | 3.131 | True | DeltaDTM_v1_1_N47W003.tif | id_obs_le_crouesty_le_crouesty_france_re | 47.5422 | -2.893 | 0.199 | True | included_match_le6km | The6kmruleisapre-specifiedextractionscreen,notanalongg-coastrepresenta | Loire-Bay-of-Biscay | B1oo_FRBo1 | B150o_FRBo1 | B250o_FRBo1 | reviewed-coastline-v2 | Loire-Bay-of-Biscay | French-Atlantic-and-Brittany | True | False | False | not_determined_from_coordinate_review | True | coordinate_plausible_not_hydrodynamic_equivalence | coordinateconnectorinspection;sourcestationidentifierwhererenamed;ex |
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| harwich-po22-uk | Harwich Po22 Uk | harwich-po22-uk-bodc.csv | 51.947 | 1.284 | 10 | 0.691 | 0.140 | harwich_po22_uk.7z | True | False | True | True | True | False | exclude d: fewer than 25 metadata years | 0.477 | 3.861 | True | Delta DTM_v1_1_N51E001.tif | id_obs_harwich_po22_uk_bodc | 51.951 | 1.282 | 0.477 | True | included_match_le6km | The 6km rule is a pre-specified extraction screen, not an along-coast representa | Southern-North Sea-UK | B1oo_UKSEo2 | B150_UKSEo2 | B250_UKSEo1 | reviewed-coastline-v2 | Southern-North Sea-UK | British-North-Sea | True | True | False | True | not_determined_from_coordinate_review | True | coordinate_pair_plausible_not_hydrodynamic_equivalence | coordinate connection inspection; source-station identifier where renamed; ex |
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Supplementary Data provenance

Machine-readable source files for all tables are retained under `data/figure_source/`. The scientific-integrity review records current hashes, invariants and known external-deposit placeholders. References are listed in the main manuscript.